

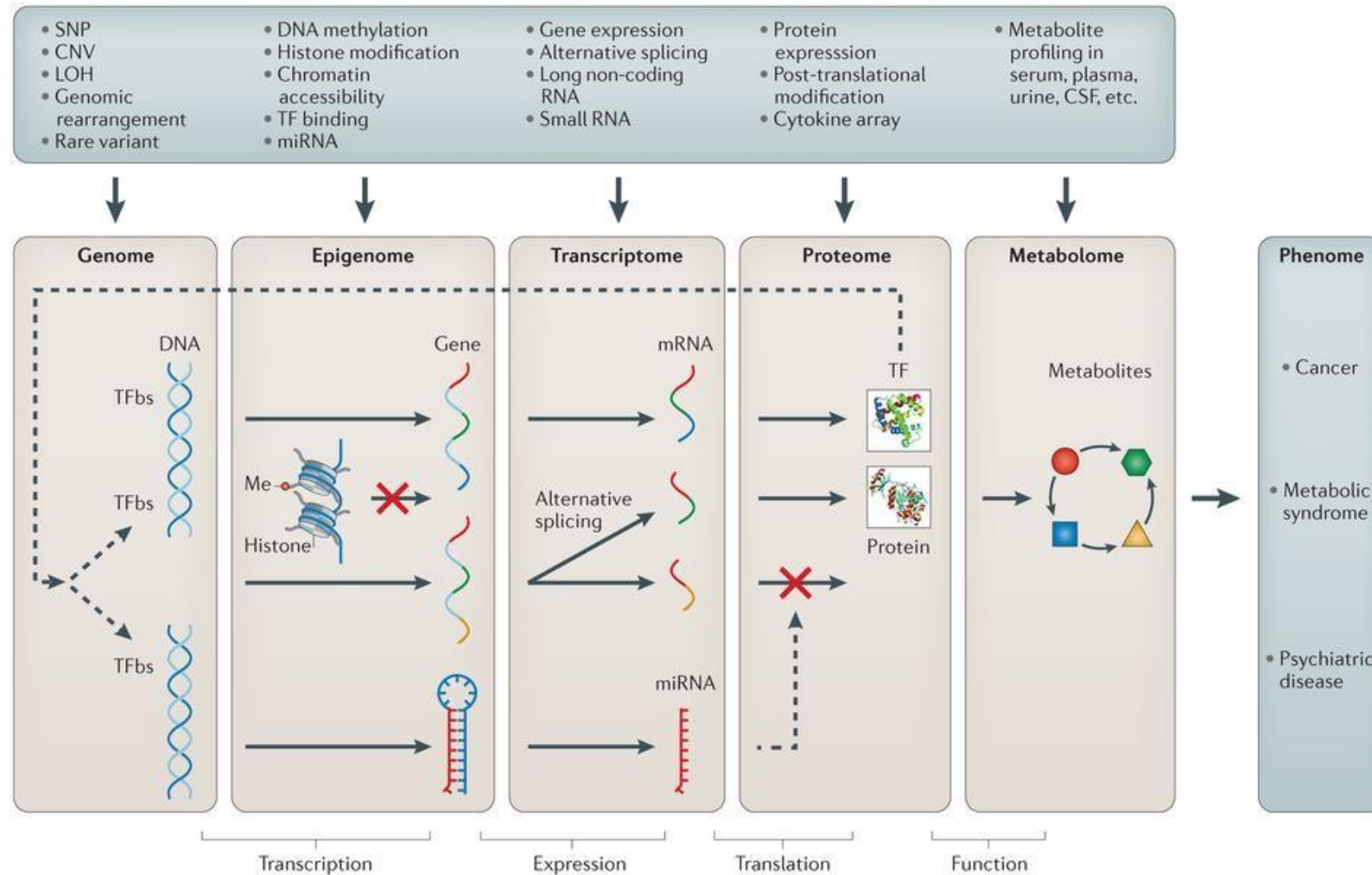
Building computational models in Systems Biology

Computational Systems Biology for Digital Medicine
7–12 December 2025
Wellcome Genome Campus, UK
Instructor: Anna Niarakis



wellcome
connecting
science

Organisms: Complex systems

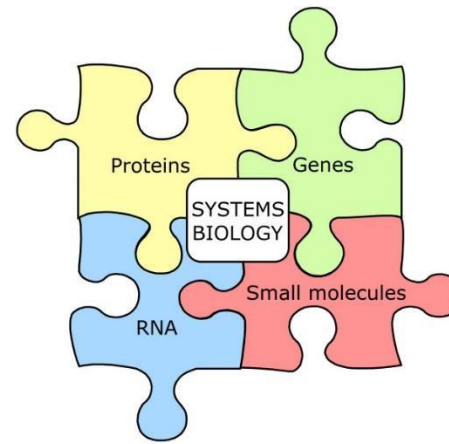


Systems biology: put bits and pieces together

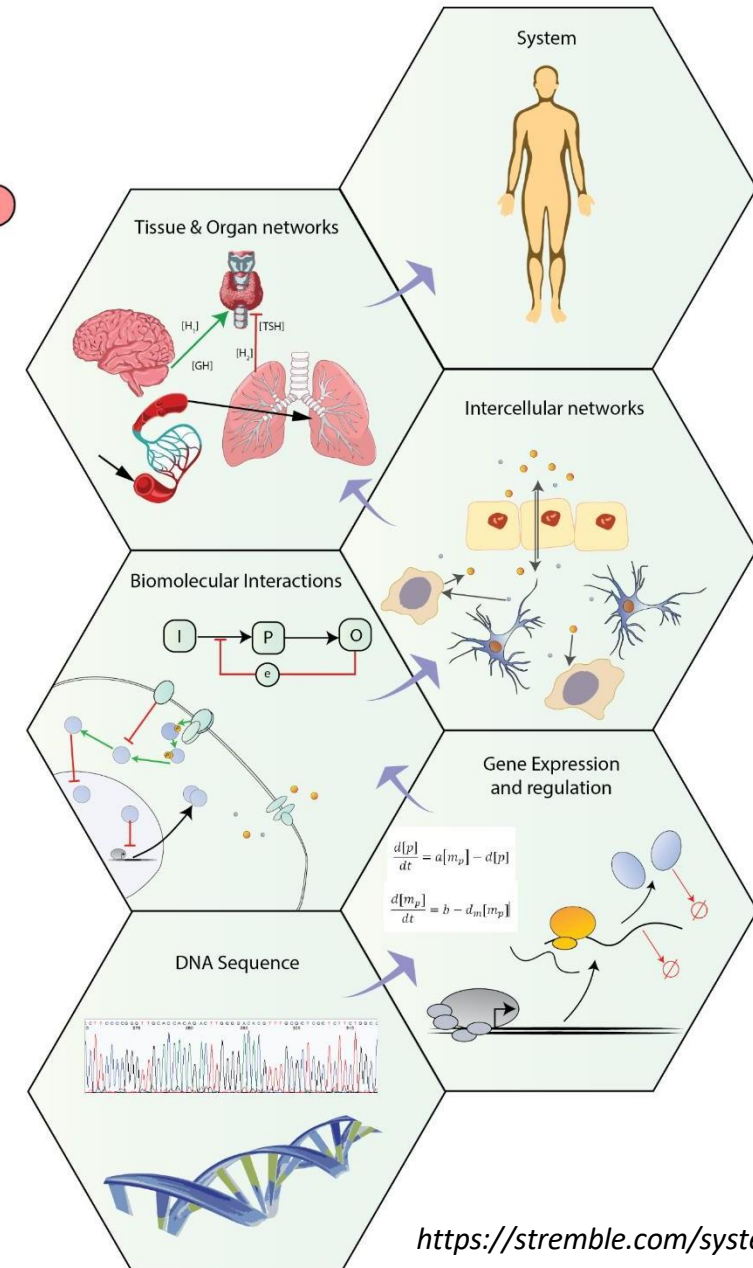
How do the individual parts interact to yield system behavior?

Biology has focused on figuring out the pieces.

But what happens when you fit them together?

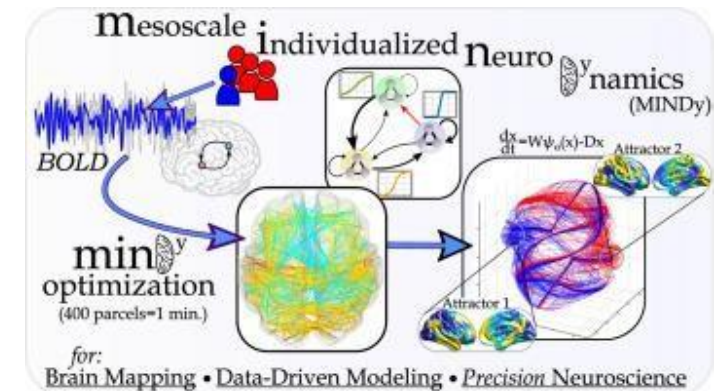
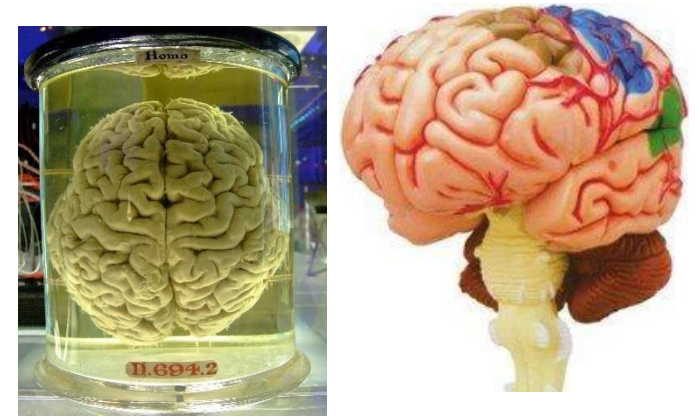


The Systems Biology Framework
Layers of Abstraction

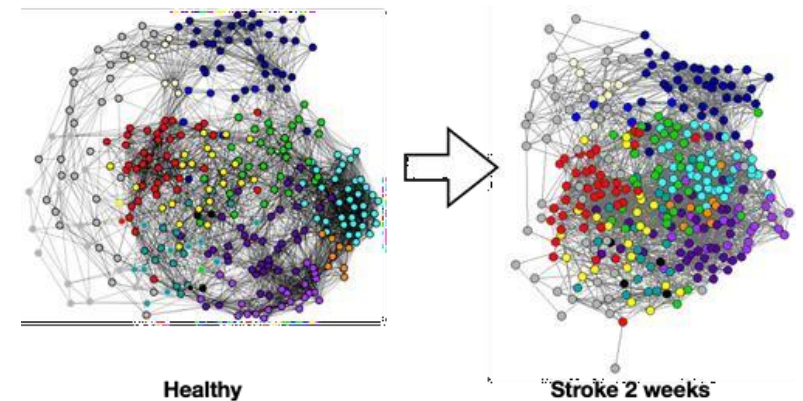


The concept of model in science

- A physical, mathematical, graphical, logical, conceptual, computational representation of an object, event or process or a system of objects, events or processes.
- Abstraction of the « real thing ».
- Helps us study and understand the **system** of interest and the **mechanisms** under which it operates.
- Can also lead to predictions about the system's behaviour.



<https://doi.org/10.1016/j.neuroimage.2020.117046>



<https://www.humanbrainproject.eu/en/>

Full scale integration

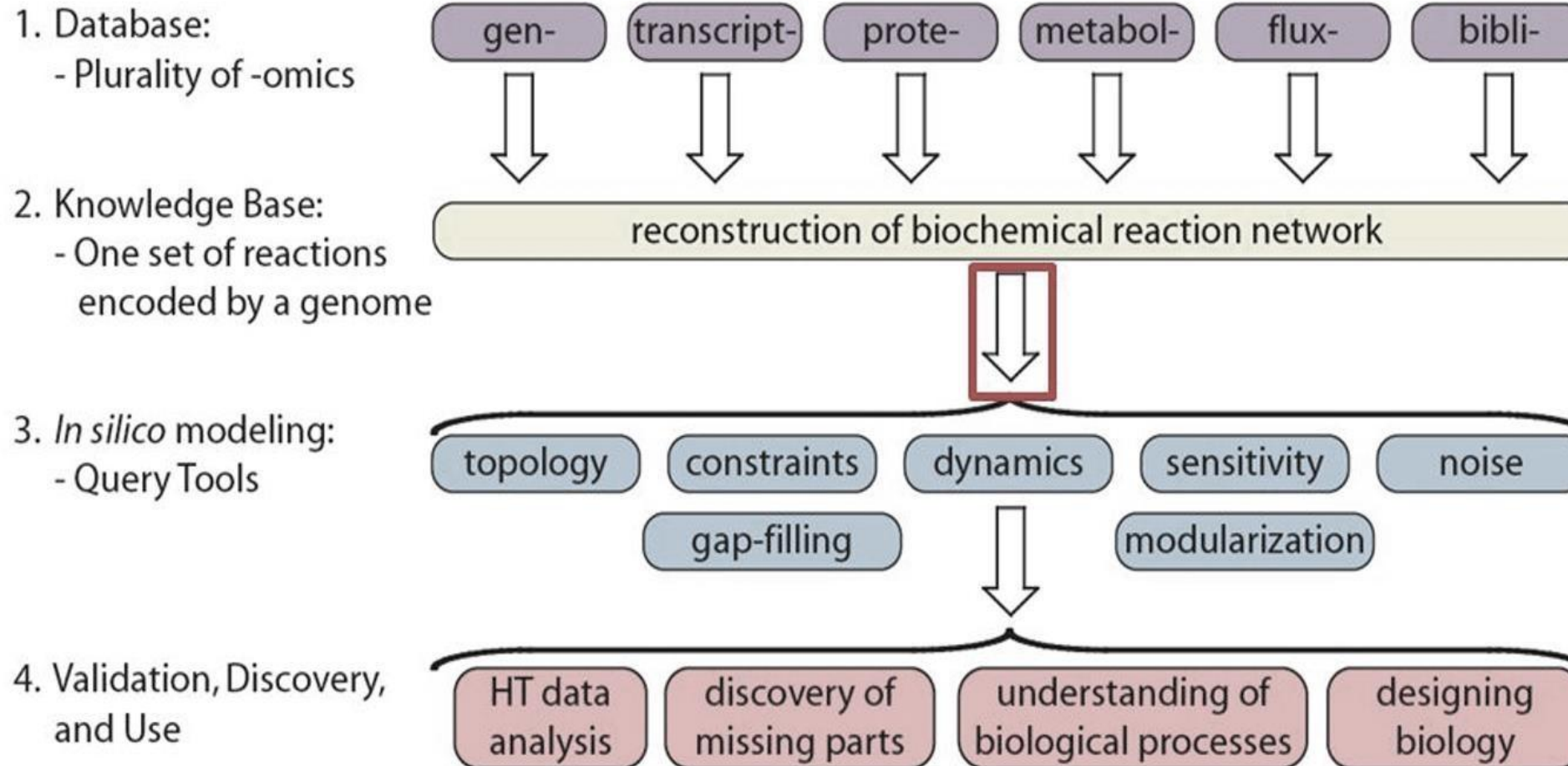
- Key goal of systems biology: construct networks at different cellular levels to investigate cellular machinery.
- Currently no satisfactory method to construct an integrated cellular network.

The challenge

- Interpret largescale data sets and extract true information to understand biological systems.
- Computational techniques, which can integrate and combine these large and heterogeneous data sets, will help gain more biological insights.

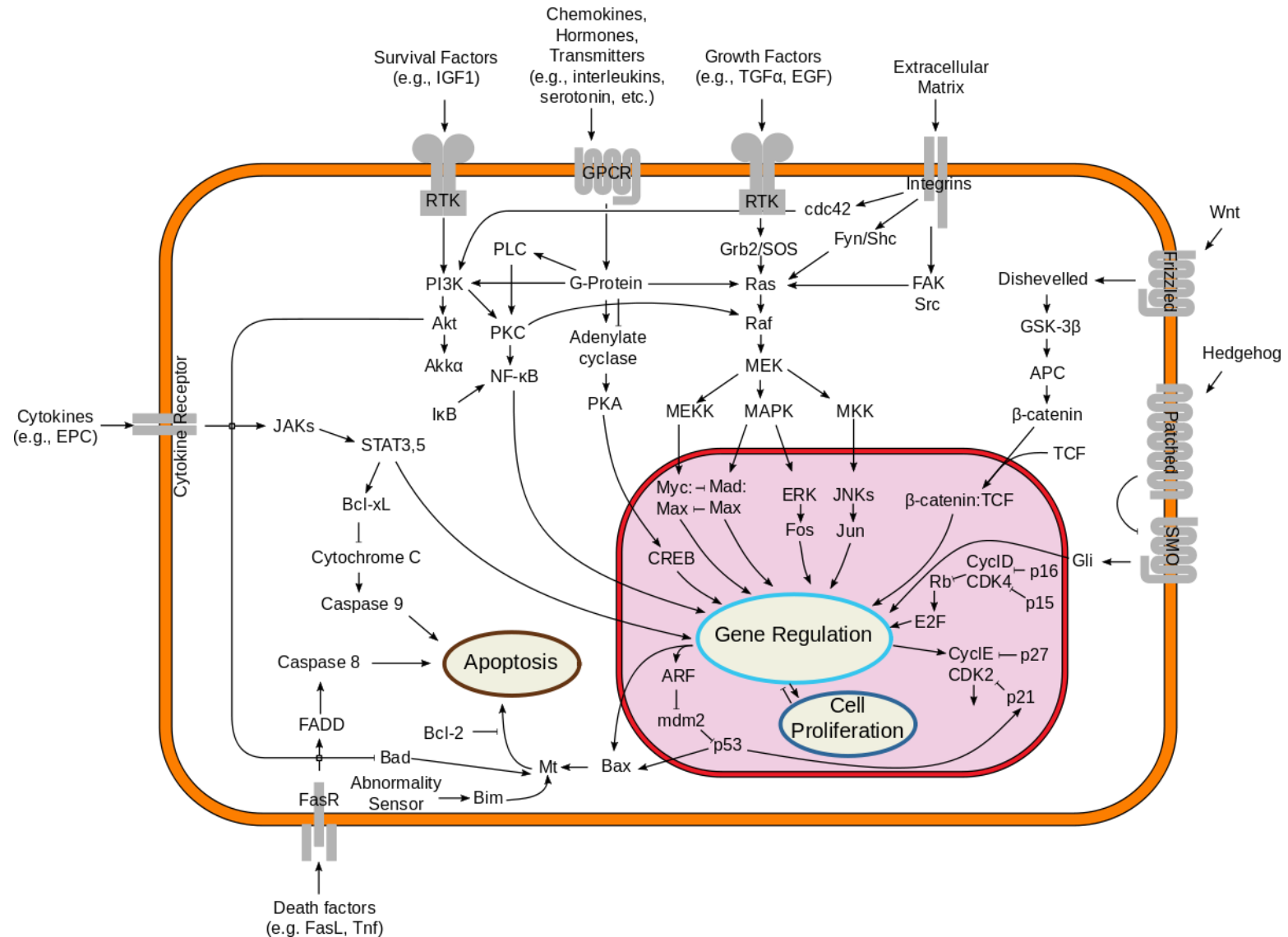
The Systems Biology paradigm:

Components -> networks -> computational models -> phenotypes



Viewing cells in terms of their underlying network

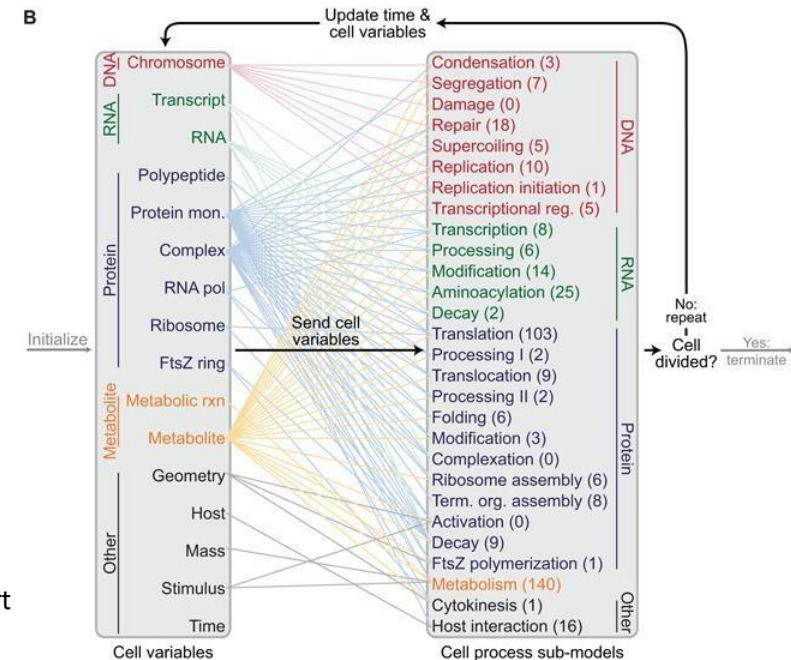
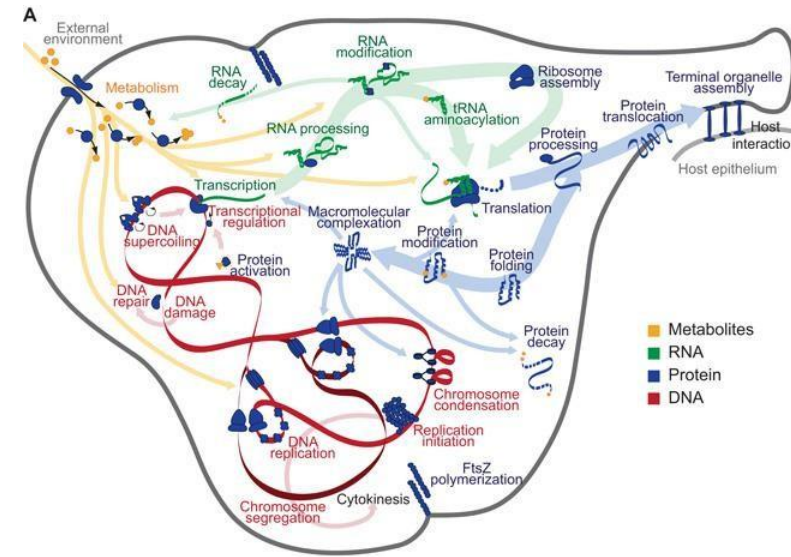
Organizing biological information in the context of networks is a powerful concept, networks can be seen as graphs.



Whole cell model

- Mycoplasma genitalium
- Computational Simulation and Analysis
- All simulations were performed with MATLAB R2010b on a 128 core Linux cluster. The predicted dynamics of each cell was logged at each time point and subsequently analyzed using MATLAB.

Karr JR, Sanghvi JC, Macklin DN, Gutschow MV, Jacobs JM, Bolival B Jr, Assad-Garcia N, Glass JI, Covert MW. A whole-cell computational model predicts phenotype from genotype. Cell. 2012 Jul 20;150(2):389-401. doi: 10.1016/j.cell.2012.05.044. PMID: 22817898; PMCID: PMC3413483.



The concept of a Digital Twin



Digital Twins are virtual equivalents, or twins, of physical objects.

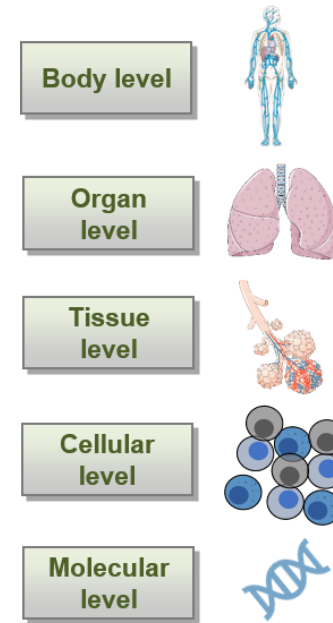
These digital copies are increasingly popular because they can be used to drive important simulations that haven't been possible until now.

A digital twin is a virtual representation that serves as the real-time digital counterpart of a physical object or process. *Image Credit: Noria Corporation*

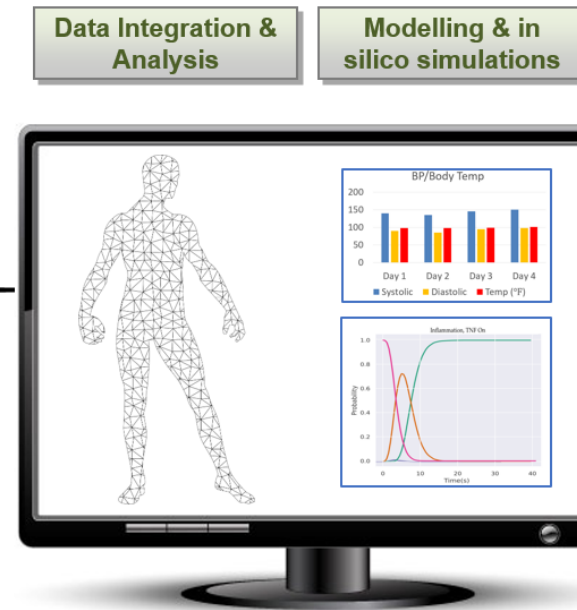
IDT requires advances in three critical areas

1. Develop sufficiently powerful and detailed simulation models of the human immune system.
2. Acquire the appropriate real-world data to train and validate the simulation models.
3. Personalize and customize the simulation models to individual patients and for particular purposes.

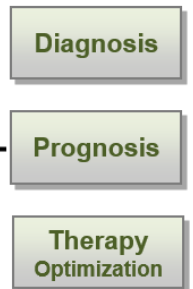
Multi-scale biological data



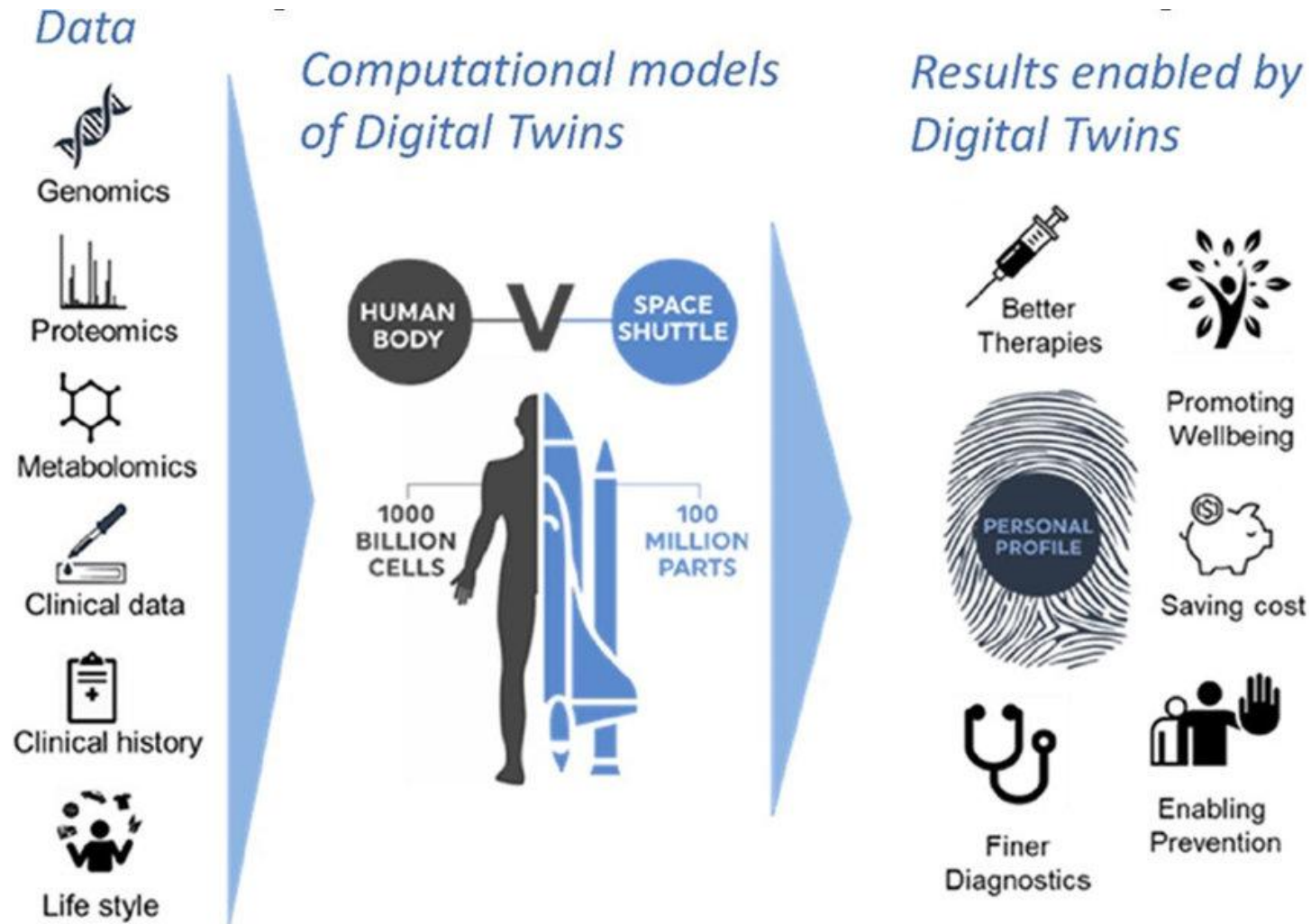
Immune Digital Twin



Personalized care



The concept of a Digital Twin for complex human



Lee et al., 2020

Human body:

- 37.2 trillion cells
- 200 different cell types

Is everything digital a “Digital Twin” ?

Digital Model:

a **virtual representation of a physical object, system, or process** (algorithm, mathematical model, 3D representation etc).

Visualization, analysis, and manipulation of objects or systems in a digital environment, aiding in design, optimisation, and testing.

Used for *in silico* simulations and hypothesis testing (predictions) of how a physical object, system, or process might operate in the future or in a particular environment.

• Digital Shadow:

- A digital shadow is an **evolving digital representation** that mirrors the current state and behaviour of a physical entity or system.
- It collects data through sensors, Internet of Things (IoT) devices, or other sources and provides a feed of information that is fed into the model.
- Typically, digital shadows are mathematical models, but they could also be 3D representations.
- They enable monitoring, predictive analysis, and decision-making.

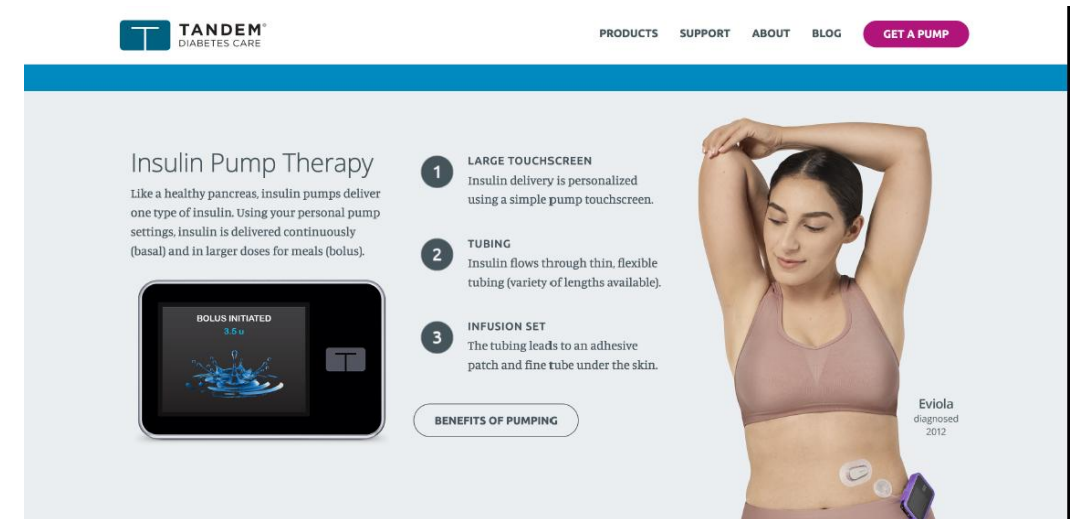
Real “Digital Twin”

Real-time connection between the physical entity and its digital counterpart, where the physical object gives information to the digital replica and vice versa.

Digital twins simulate, monitor, and **control** physical objects or systems, facilitating analysis, optimisation, and predictive maintenance.

They enable **live feedback loops and foster insights for improving performance, efficiency, and reliability.**

To put it differently, there is a two-way interaction between the physical and the digital environments, where the digital replica is able to change how the physical entity operates.



HELPS PROTECT YOU FROM LOWS

Basal-IQ® predictive low-glucose suspend technology on the t:slim X2™ insulin pump.



HELPS PROTECT YOU FROM HIGHS AND LOWS

Control-IQ™ advanced hybrid closed-loop technology on the t:slim X2 insulin pump.



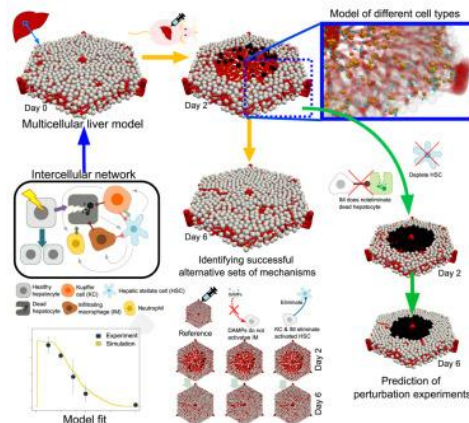
Significant efforts

Cardiovascular diseases (<https://www.simcardiotest.eu/wordpress/project/use-cases/>)

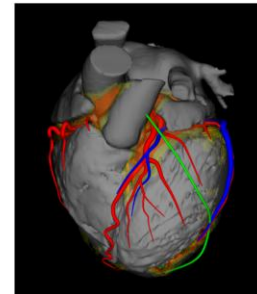
Virtual Liver

Liver Systems Medicine

striving to develop non-invasive methods for diagnosing and treating NAFLD by combining mathematical modeling and biological research.



PACING DEVICES AND LEADS

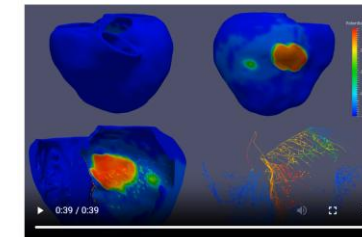


Coronary and cardiac structure, @IHU Lyric

The first case on pacing devices will set up case-studies for in-silico simulation of catheterisation, implantation and pacing of cardiac devices, as a potential support to their development or certification. These use cases will cover the catheter navigation, the pacing performance and the mechanical risks, including perforation and fatigue. It will leverage a physical simulator using 3D printing, soft robotics, experimental data acquisition, and computer simulations in order to help the design and validation of such devices. We will make use of computational tools and expertise developed at [Sofa](#) (Simulation open Framework Architecture), and the Cardiac Electrophysiology Solver (CEPS) and use industrial catheters and pacing devices. The aim is to develop computational models of the navigation approaches, the computational evaluation of the pacing properties, and ultimately mechanical interaction (i.e., more specifically on perforation and fatigue). This will impact the design and the material on mechanical interactions, and will provide the best practical threshold and impedance for electrical performance.

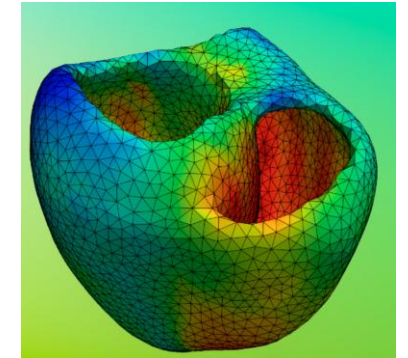
[READ MORE](#)

THE DRUGS, EFFICACY AND TOXICITY

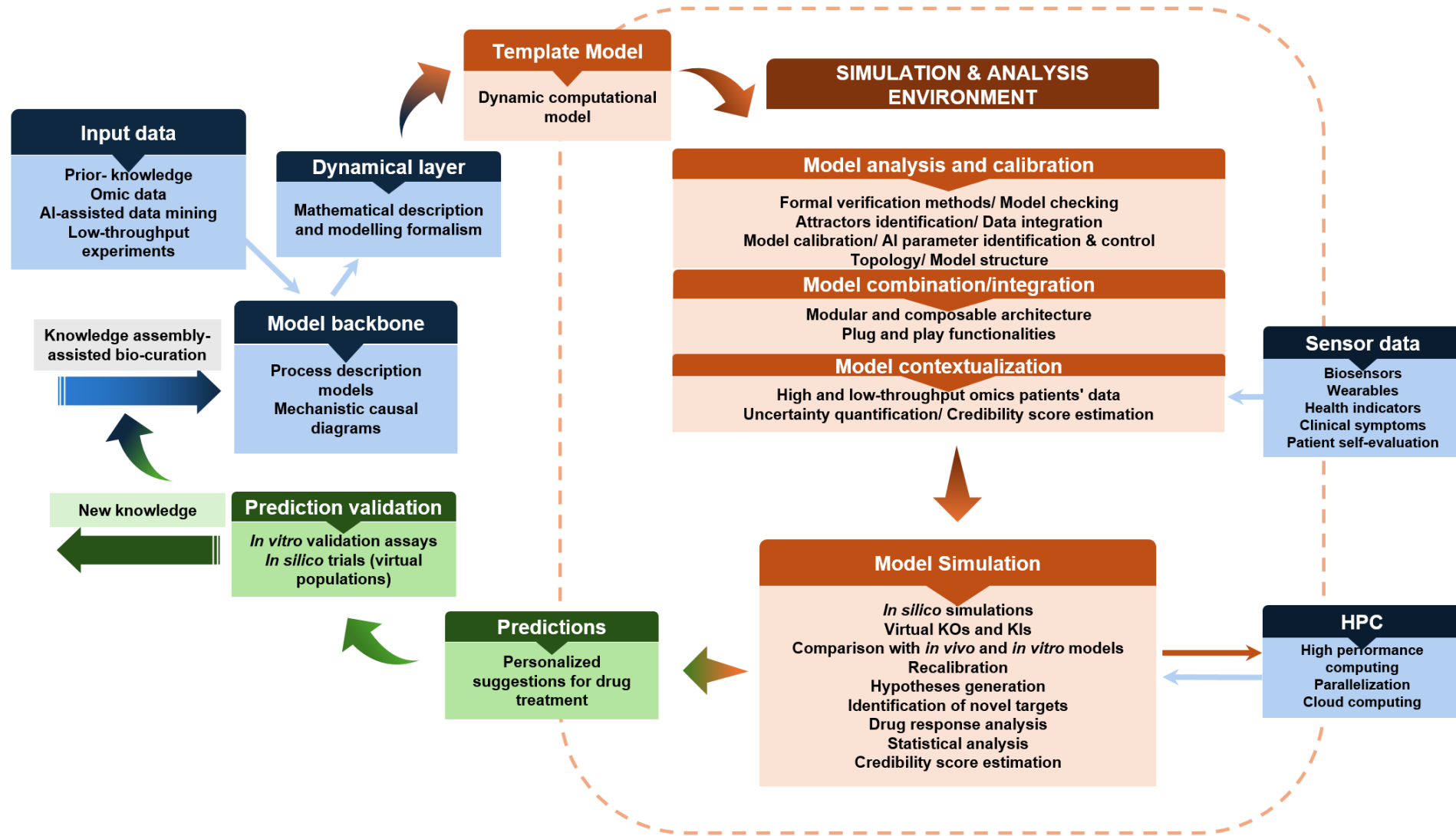


The **Drugs, efficacy and toxicity** will develop models for drug-ion channel interaction taking into consideration the plasmatic concentrations derived from pharmacokinetic and pharmacodynamic studies. The models will be used to analyse the effects of drugs at a multi-scale level from cell to organ, in normal and pathological conditions, such as ischemia, heart failure or atrial fibrillation. Personalised models will be built according to standardisation process, populations of models where variability will be considered, as well as differences in gender and age. Specific biomarkers to assess drug efficacy and safety will be defined.

[READ MORE](#)

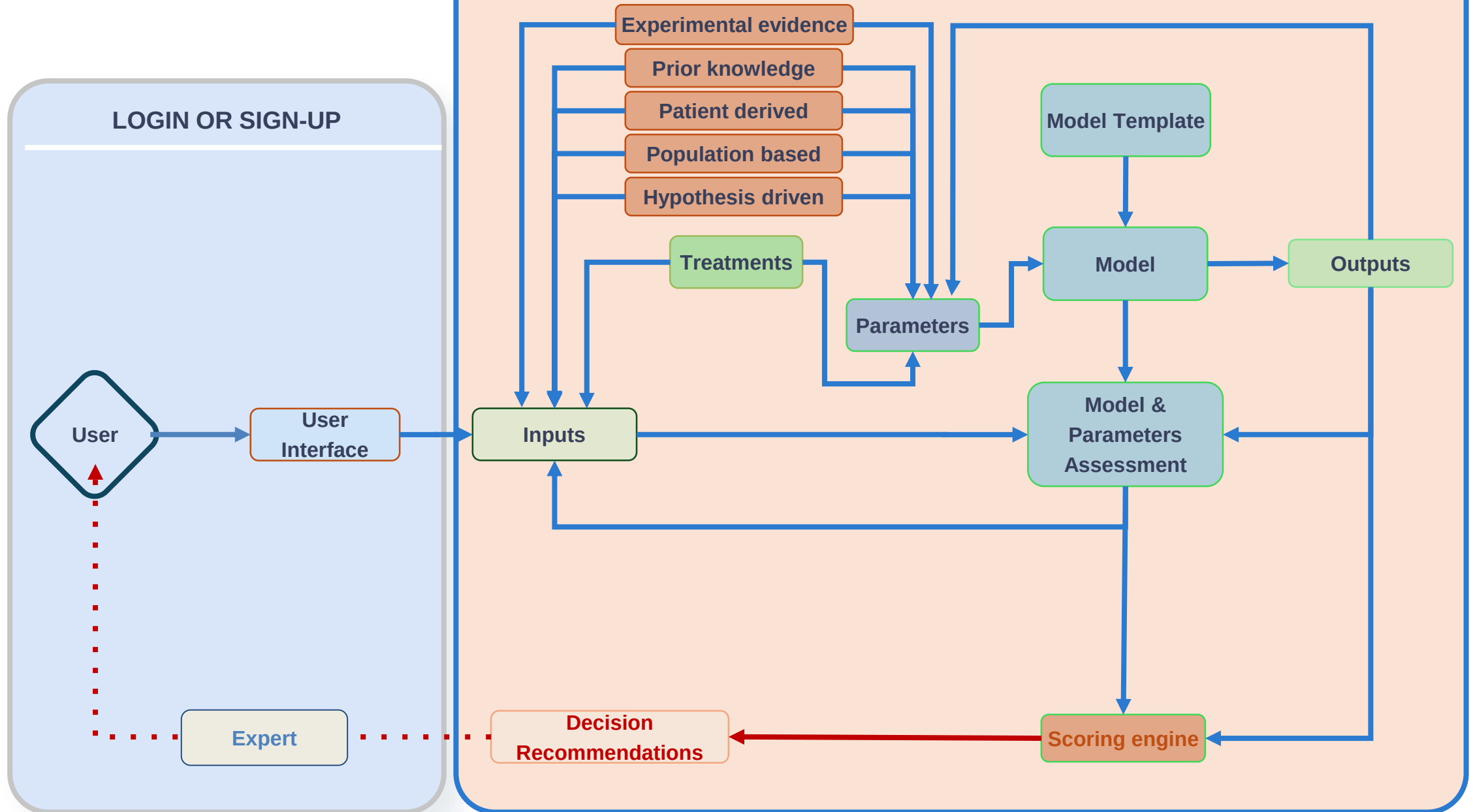


Digital Twins: more than a model

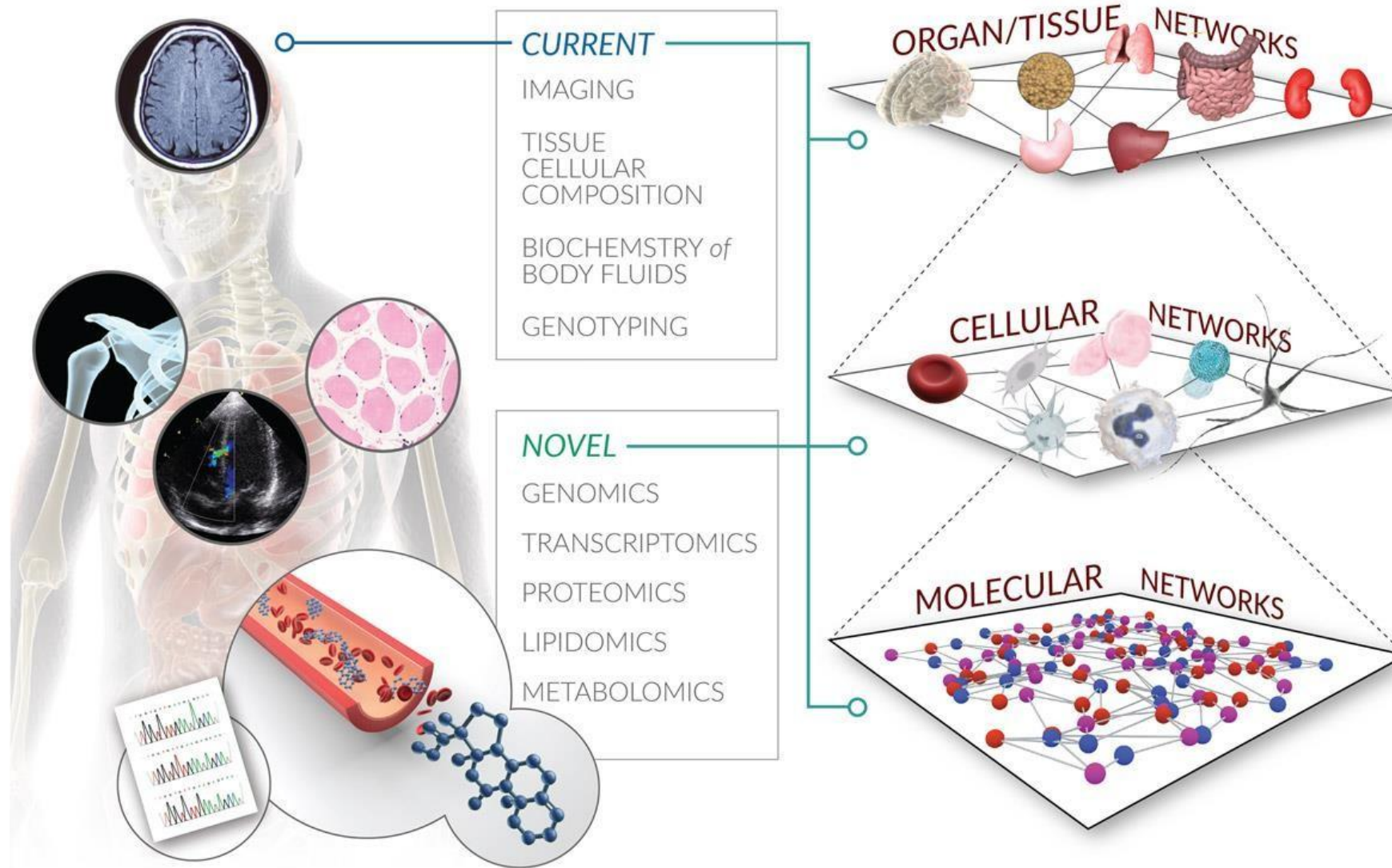


IDT CONCEPTUAL DESIGN

LOGIN OR SIGN-UP

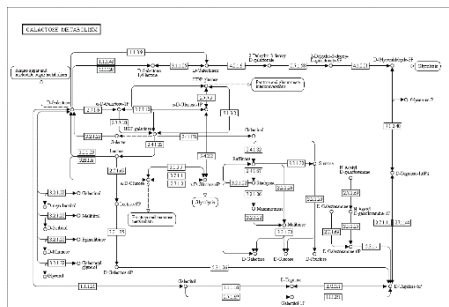
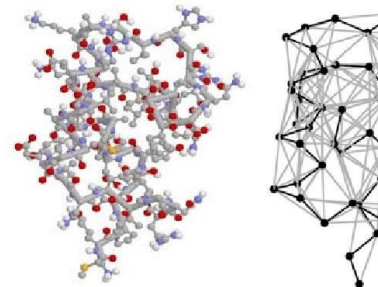
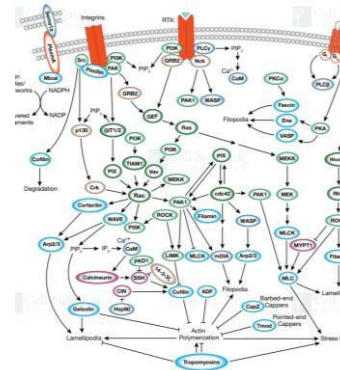
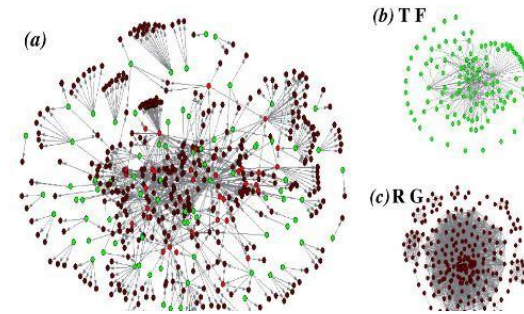
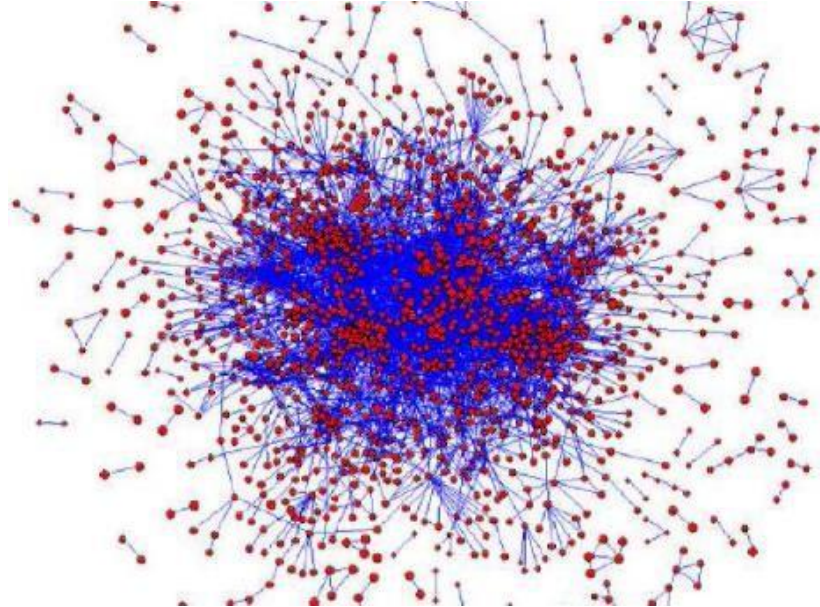


DIAGNOSTIC APPROACHES



Intra-cellular networks

- Transcriptional regulation networks
- Protein structure networks
- Metabolic networks
- Protein-protein interaction (PPI) networks
- Cell signaling networks



How do we construct a network?



- **Bottom up (from scratch)**
 - Literature based
 - Curation
 - Databases
 - Previous Knowledge
 - From local to global
- **Top down (data driven)**
 - Data dependent
 - Algorithms
 - Inference
 - Reverse engineering

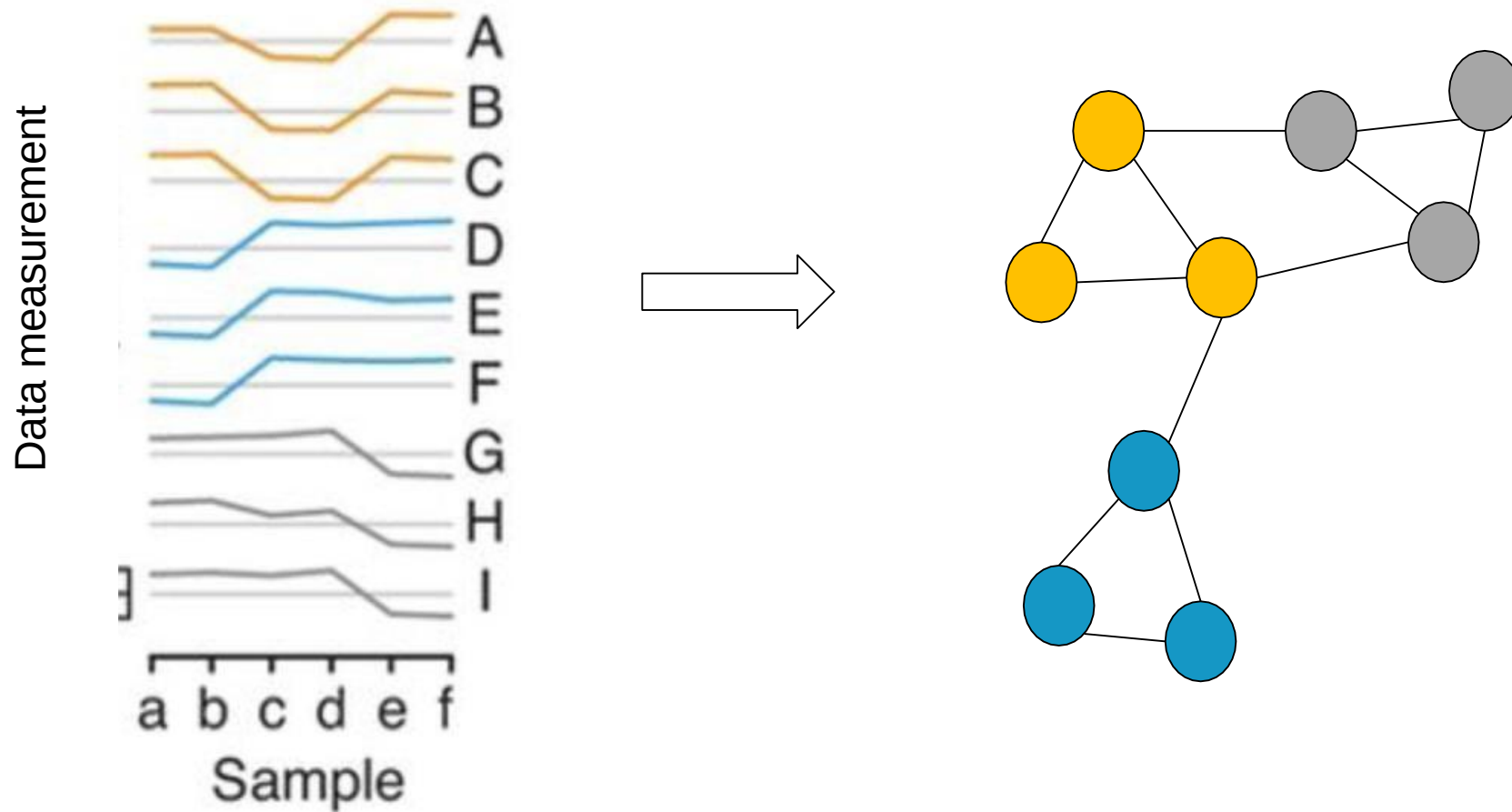


Top down
Data driven network inference

What is network inference?

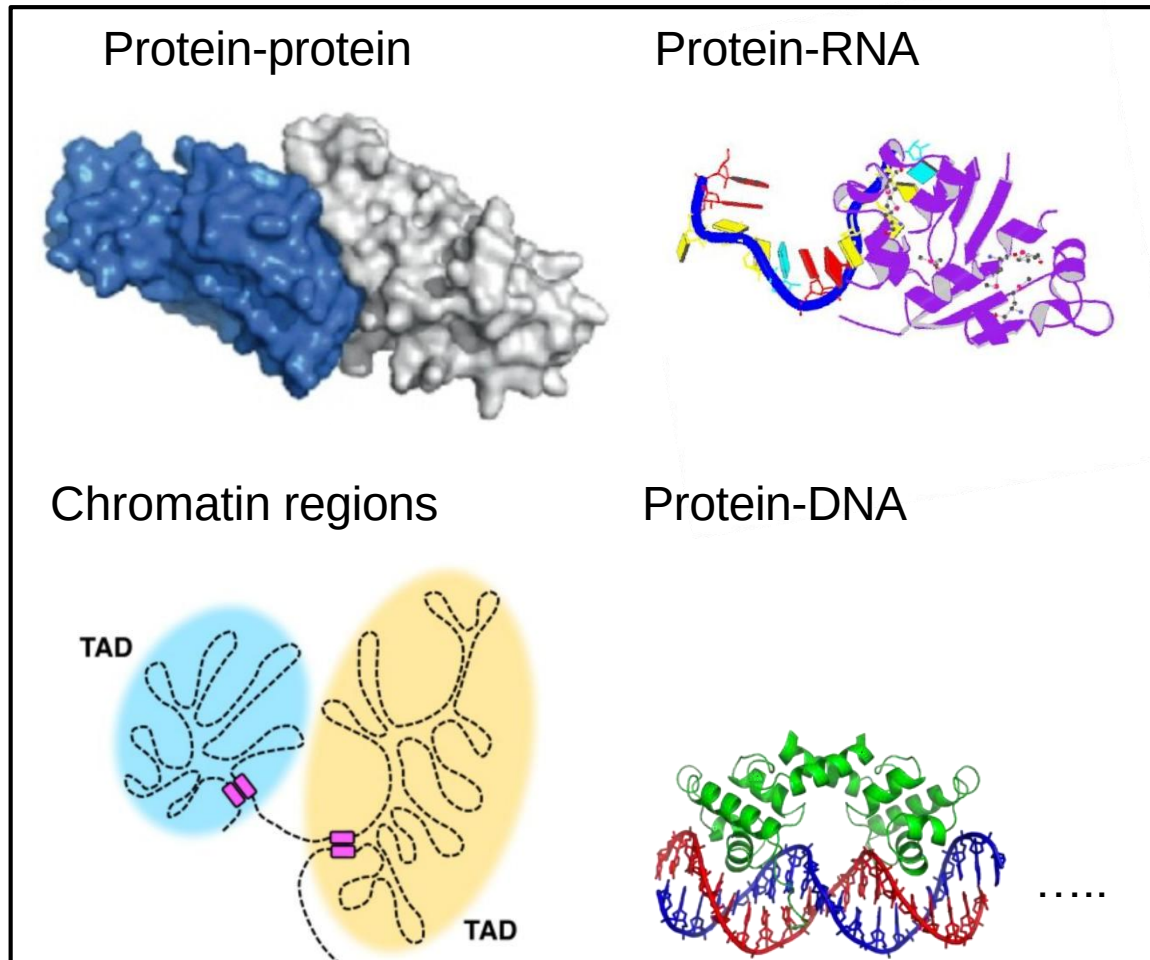
"process of revealing the network structure of a biological system by reasoning backward from observed data."

He et al, 2009



Why do we need network inference?

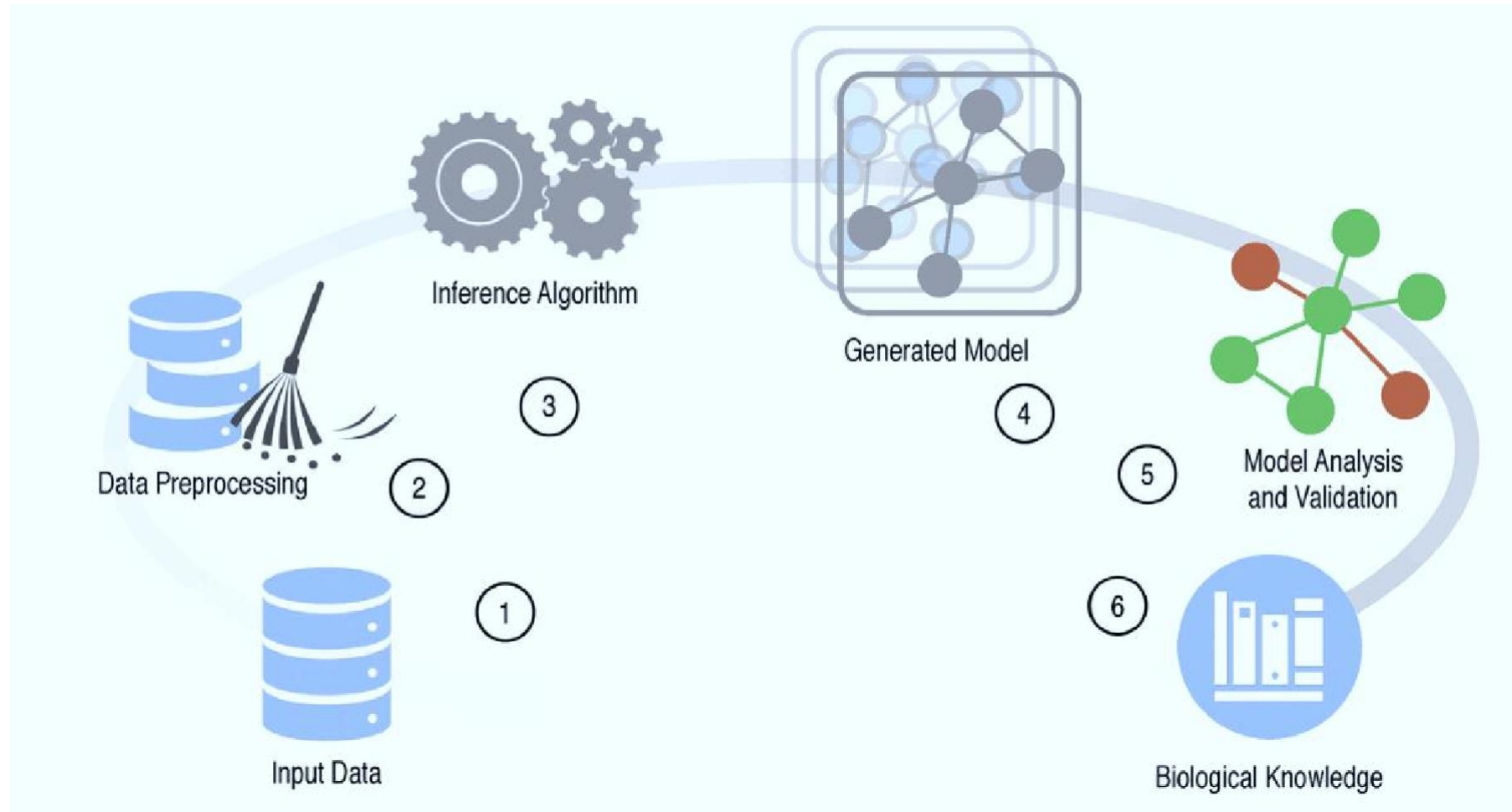
Most/All functions in cells are performed through interactions



Interactions are hard to measure

Abundances are easier to measure

Network inference process



Strengths/weaknesses of network inference types

Computational approach	Strengths	Weaknesses
Information-theory models	• Large GRNs, even out of low expression genes • Mutual and conditional mutual information approach • Not computationally-demanding • Low number of samples	• Regulation by multiple genes is not considered • Static, only suitable for steady-state data
Examples:	REVEAL [36], RELEVANCE [40,84], ARACNE [42], CLR [43], MRNET [44]	
Boolean models	• Capable of inferring large networks • Generally easy to interpret • Simplify underlying complex biological phenomena • Allow supervised learning methods	• Deterministic nature • Discretization bottleneck (only on/off states) • Problems in handling incomplete or inconsistent expression data • High computing time • Most of them use small number of genes
Examples:	RCGA [85], TRaCE + [86], CABERNET [87]	
ODE models	• Directed signed graphs • Realistic dynamics • Suitable for both steady-state and time series expression data • Simplification of the system by means of linear functions • Allow prediction of the behaviour of the network under different conditions once parameters are known	• Not suitable for large networks • Linear functions also constrain the dynamic behaviour of cell regulatory functions (e.g. oscillations, multistationarity) • Hard to find appropriate values for model parameters • Noisy data leads to qualitative instead of quantitative GRN inference
Examples:	SCODE [68], HiDi [69]	
Bayesian models	• Noise and uncertainty handling • Do not require a large number of involved variables • Integration of prior knowledge and allowance of enrichment analyses • Statistical inference of gene network	• Feedback loops are not allowed • Fail in the inference using time series expression data • Cannot cope with large GRNs • Inherent combinatorial learning
Examples:	F-MAP [88], MDP [89], POMDP [71], QMR-DT [73]	
Neural models	• Recognize an input pattern • Model any functional relationship inferable from the data • Suitable for both steady-state and time series expression profiles • Noise handling and biologically plausible • Manage non-linear and dynamic behaviour	• Machine training experiments are hard to perform since every situation requires a different learning rate definition • Computational complexity makes them more suitable for very small systems
Examples:	ANN [27], RNN [78], ELM [83]	

Considerations when choosing approach

1. Simplicity/interpretability
2. Scalability
3. Assumptions (e.g. can we assume linear relationship?)
4. Sensitivity vs specificity
5. Ability to model feedback loops and combinatorial regulations
6. Availability of data: how much, statistical independency, data points, etc
7. Pitfalls of different network types: e.g. from mRNA to gene regulatory networks:
mRNA abundance is a proxy for activity and presence of regulators, not a direct influence
8. Integrating prior knowledge improves results

Considerations for experimental design / input data

- Network inference is a largely undetermined problem. i.e. has multiple solutions
- To help deconvolute uncertainties, the best experimental design includes
 - Perturbation data
 - Steady-state (more realistic as cheaper) or time course (ideal)
 - Timing for measurements is important
- Bigger scale of network to infer requires more data
- Replicates are needed to remove stochastic error and ensure quality of data

Optimization of networks

- Dimensionality reduction can help focus on ‘interesting’ features/genes
 - E.g. Feature selection: remove genes with low expression or no changes
- Structure/topology optimization against experimental datasets
 - Forward selection (growing network and evaluating)
 - Backward elimination (pruning and evaluating)
- Integration of prior knowledge has been shown to almost always increase accuracy
- Integration of other datasets also has been promising but tools are in early development stage



Bottom up network construction

		Metabolic	Protein- protein	Regulatory/ Signaling	Organisms	Curation ^a
KEGG	http://www.genome.jp/kegg/	x			many	C
BiGG	http://bigg.ucsd.edu/	x			many	M
BioCyc ^b	http://biocyc.org/	x		x	many	C/M
MetaCyc	http://metacyc.org/	x			many	C/M
Reactome	http://reactome.org/	x	x	x	many	M
BIND	http://www.bindingdb.org/		x		many	E/M
DIP	http://dip.doe-mbi.ucla.edu/		x		many	M
HPRD	http://www.hprd.org/		x		human	M
MINT	http://mint.bio.uniroma2.it/		x		many	M
Biogrid	http://www.thebiogrid.org/		x		many	E
UniHI	http://theoderich.fb3.mdc-berlin.de:8080/unihi/		x		human	E/M
Yeasttract	http://www.yeasttract.com/			x	yeast	M
TRANSFAC	http://www.gene-regulation.com			x	many	M
TRANSPATH	http://www.gene-regulation.com			x	many	M
RegulonDB	http://regulondb.ccg.unam.mx/			x	many	C/E
NetPath	http://www.netpath.org/			x	human	M

^aM=Manual/Literature, C= Computational, E= Experimental.

^blinks to other *Cyc databases

Many sources available

Database Name	URL or Web Address	Comments
HumanCyc (Encyclopedia of Human Metabolic Pathways)	http://humancyc.org/	-MetaCyc adopted to human metabolism -No disease or drug pathways
KEGG (Kyoto Encyclopedia of Genes and Genomes)	http://www.genome.jp/kegg/	-Best known and among the most complete metabolic pathway databases -Covers many organisms -A Few disease and drug pathways
The Medical Biochemistry Page	http://themedicalbiochemistripage.org/	-Simple metabolic pathway diagrams with extensive explanations -A few drug and disease pathways
MetaCyc (Encyclopedia of Metabolic Pathways)	http://metacyc.org/	-Similar to KEGG in coverage, but different emphasis -Well referenced -No disease or drug pathways
Reactome (A Curated Knowledgebase of Pathways)	http://www.reactome.org/	-Pathway database with more advanced query features -Not as complete as KEGG or MetaCyc
Roche Applied Sciences Biochemical Pathways Chart	http://www.expasy.org/cgi-bin/search-biochem-index	-The old metabolism standard (on line) -Describes most human metabolism
Small Molecule Pathway Database (SMPDB)	http://www.smpdb.ca/	-Pathway database with disease, drug and metabolic pathways for humans -Extensive search, analysis and visualization tools
Wikipathways	http://www.wikipathways.org	-Community annotated pathway database for 19 model organisms -Contains 175 human pathways -Few drug or disease pathways

	CBN	KEGG	Reactome	BioCarta	Wiki-pathways	SPIKE	UCSD signaling gateway	NCI pathway interaction database	NetPath
Species [human (Hs); mouse (Mm); rat (Rn)]	Hs, Mm*, Rn*	>20 species	Hs (curated) +20 species (inferred)	Hs, Mm	>25 species	Hs	Hs, Mm	Hs	Hs
Literature support displayed									
At edge level	✓					✓	✓	✓	✓
At pathway level		✓	✓	✓	✓			✓	
Defined biological boundaries									
Species	✓	✓		✓	✓	✓		✓	✓
Tissue	✓								
Disease context	✓	✓			✓				✓
Biological pathways	✓	✓	✓	✓	✓	✓		✓	✓
Manual curation	✓	✓	✓	✓	✓	✓	✓	✓	✓
Data-driven enhancement	✓						✓		✓
Crowd curation	✓			✓	✓				±
Directional edges	✓	✓	✓	✓	✓	✓	✓	✓	✓
Multiple types of entities	✓	✓	✓	✓	✓	✓	✓	✓	
Interactive visualization	✓		✓		✓				
Computable	✓	✓	✓						
Available for download	✓	✓	✓		✓	✓	✓	✓	✓
Size	>120 network models	>450 pathway maps	>1400 pathways (Hs)	>350 pathways	>430 pathways	>25 curated pathways	~3500 proteins and their proximal connections	>135 NCI-Nature curated pathways (+Reactome + BioCarta)	>30 curated pathways (immune signaling/ cancer)

*coming soon

Causal biological network database: a comprehensive platform of causal biological network models focused on the pulmonary and vascular systems

Pathway-bases are now enriched in tools and functionalities



reactome.org

☆

🔍

📄

📁

📄

📄

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reactome

AboutContentDocsToolsCommunityDownload

Find Reactions, Proteins and Pathways

g. O95631, NTN1, signaling by EGFR, glucose

Go!

Pathway Browser

Visualize and interact with Reactome biological pathways

Analysis Tools

Merges pathway identifier mapping, over-representation, and expression analysis

ReactomeFIViz

Designed to find pathways and network patterns related to cancer and other types of diseases

Documentation

Information to browse the database and use its principal tools for data analysis

Analysis tools

⌵

Analyse gene list

Analyse gene expression

Species Comparison

Tissue Distribution

Your dataOptionsAnalysis

Step 1: Select a file from your computer or paste your own data and click on the corresponding "Continue" button.

Select data file for analysis: Choisir un fichierAucun fichier choisi

Continue

Paste your data to analyse or try example data sets:

Paste your data here or select an example from the right >>

Some examples:

UniProt accession list

Gene name list

Gene NCBI / Entrez list

Small molecules (ChEBI)

Small molecules (KEGG)

Microarray data

Metabolomics data

Cancer Gene Census (COSMIC)

Tissue Specific Expression (HPA)

Clear

Continue

←→↺

genome.jp/kegg/network.html

KEGG

Databases

Tools

Auto annotation

Kanehisa Lab

KEGG

Network

KEGG NETWORK Database

From gene variation to network variation

KEGG2PATHWAYBRITEMODULEKONETWORKDISEASEDRUGMEDICUS

Search NETWORK in KEGG MEDICUS

rheumatoid arthritis

Go

Search human gene VARIANT in KEGG MEDICUS

Go

Network Variation Maps

KEGG NETWORK represents a renewed attempt by KEGG to capture knowledge of diseases and drugs in terms of perturbed molecular networks (see Background of KEGG DISEASE). It accumulates variations of molecular interaction/reaction networks in terms of network variation maps (such as nt06210) consisting of network elements (such as N00014). The notation of network variation maps is as follows.

Network element	Coloring
Reference network	Green
Variant network containing	
Human gene variant	Red
Pathogen gene/protein	Purple
Environmental factor	Blue
Drug-target relation	Navy

Edge	Interaction/reaction
→	Activation
⇢	Inhibition
=	Complex formation
≠	Missing interaction or reaction
↗	Gain of function
⇒	Expression
⇐	Repression
—	Substrate binding to enzyme or transporter
→	Enzymatic reaction or transport process
⇒	Enzyme-enzyme relation of successive reactions

Pathway-bases are now enriched in tools and functionalities

← → ↺

pantherdb.org



PANTHER
Classification System

The mission of the PANTHER knowledgebase is to support biomedical and other research by providing comprehensive information about the evolution of protein-coding gene families, particularly protein phylogeny, function and genetic variation impacting that function. [Learn more](#)

PANTHER18.0 Released. [Click](#) for more details.

search keyword

All

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Home

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Data Version

Tools

API/Services

Publications

Workspace

Downloads

FAQ/Help/Tutorial

Current Release: **PANTHER 18.0** | 15,693 family

Gene List Analysis

Browse

Sequence Search

Genetic Variant Im



PANTHER
Classification System

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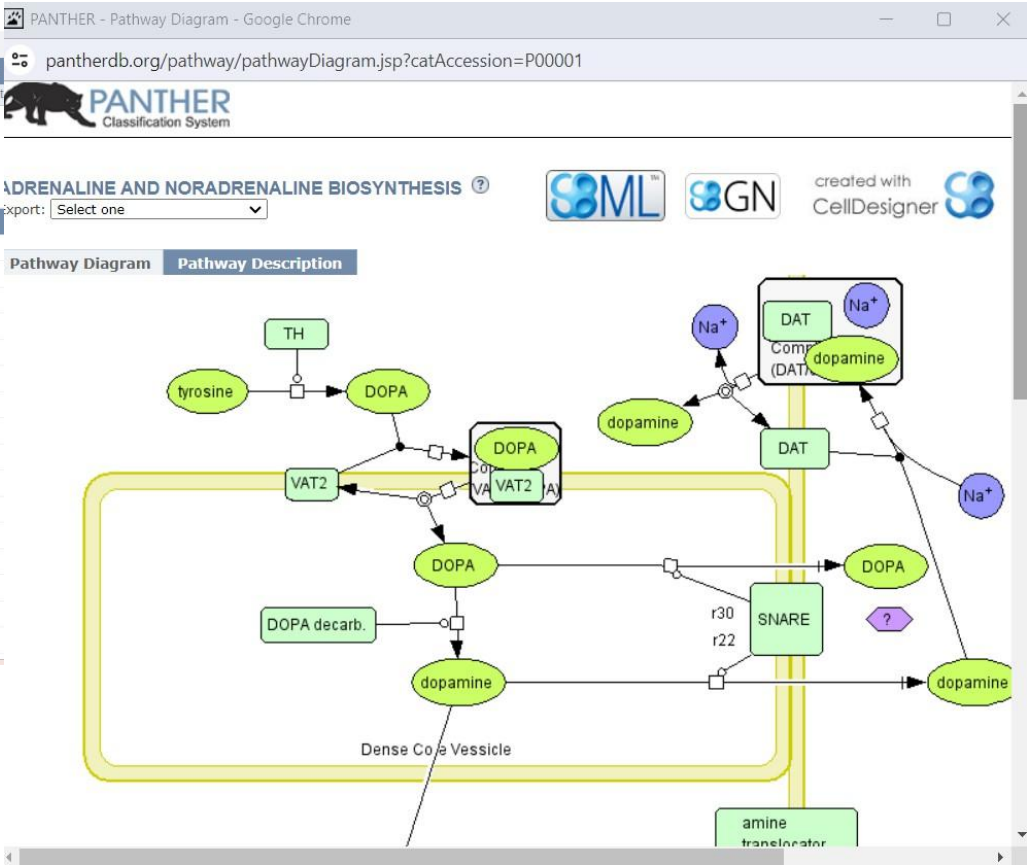
Data Version

Tools

API/Services

Publications

Community Pathway Curation | Browse Pathways | Search Pa



PANTHER PATHWAYS

- Pathway
- 2-arachidonoylglycerol biosynthesis
 - 5HT1 type receptor mediated signaling pathway
 - 5HT2 type receptor mediated signaling pathway
 - 5HT3 type receptor mediated signaling pathway
 - 5HT4 type receptor mediated signaling pathway
 - 5-Hydroxytryptamine biosynthesis
 - 5-Hydroxytryptamine degradation
 - Acetate utilization
 - Activin beta signaling pathway
 - Adenine and hypoxanthine salvage pathway
 - Adrenaline and noradrenaline biosynthesis
 - Alanine biosynthesis
 - Allantoin degradation
 - ALP23B signaling pathway
 - Alpha adrenergic receptor signaling pathway
 - Alzheimer disease-amyloid secretase pathway

Please refer to our article in [Nature Protocols](#) for detailed instructions on how to use this page.

Help Tips

Steps:

- 1. Select list and list type to analyze
 - 2. Select Organism
 - 3. Select operation
- [Using enhancer data](#)

1.

Enter ids and or select file for batch upload. Else enter ids or from workspace for comparing to a reference list.

Enter IDs:
[Supported IDs](#)

Upload IDs:
[File format](#)

Choisir un fichier

Aucun fichier choisi

separate IDs by a space or com

Please [login](#) to be able to select lists from your workspace.

Select List
Type:

- ☒ ID List
- ☐ Previously exported text search results
- ☐ Workspace list
- ☐ PANTHER Generic Mapping
- ☐ ID's from Reference Proteome Genome
- Organism for id list
- ☐ VCF File
- Flanking region
- ☐ Search Enhancer Data

2.

Select organism.

- Homo sapiens
- Mus musculus
- Rattus norvegicus
- Gallus gallus
- Danio rerio

Pathway-bases are now enriched in tools and functionalities

wikipathways.org

WikiPATHWAYS

About

Search

Browse

Communities

Download

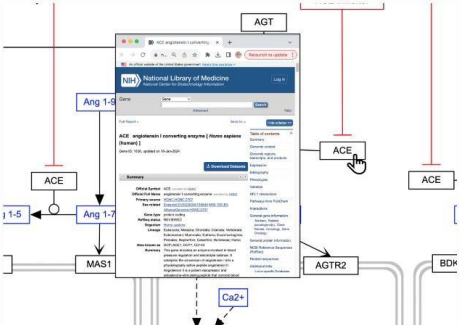
Analyze

Cite

Help

Go to Classic Site

e.g., ace2 aldosterone hu



Interact with diagrams with clickable genes, drugs, and pathways.

WikiPathways is an open science platform for biological pathways contributed, updated, and used by the research community.

Read more

Video tour

Powered by:  PathVisio &  BridgeDb

signor.uniroma2.it

The SIGNaling Network Open Resource

ABOUT

DOCUMENTATION

USER GUIDE

STATISTICS

DOWNLOADS

APIs

LOGINCONTACT

Search

Disease Browser

Pathway Browser

Advanced Methods

 Homo sapiens

 Mus musculus

 Rattus norvegicus

SEARCH

reset

all

connect

find path

add bridge proteins

Search tips:

To search a single entity type its name or ID into the search bar.

For a multi-protein search type their Uniprot IDs or Gene Names separated by one of the following delimiters: comma(,), semi-colon(;), space.

Related Projects

News

Tweets from @signor_database



COVID-19 Network Hallmarks



Nothing to see here



PathVisio

Biological pathway creation and curation software

Home

Download

Documentation

Plugins

Tutorials

FAQ

How to cite

Contact

PathVisio biological pathway editor

PathVisio is a free open-source pathway analysis and drawing software which allows drawing, editing, and analyzing biological pathways. It is developed in Java and can be extended with plugins.

PathVisio is the pathway editor for WikiPathways, a community-curated pathway database enabling collaborative pathway editing.

Download and Access

Get pathway information in the format you need, including GPML (XML), GMT, SVG, and more. Programmatically access our content in multiple ways.

GPML

GMT

SVG

Archive

API

SPARQL

R

Python



IMEx The International Molecular Exchange Consortium

Search the
Use as input: U

Home | About | Curation | Submit Your Data | Contact us

IMEx data

- A non-redundant set of physical molecular interaction data from a broad taxonomic range of organisms.
- Expertly curated from direct submissions, peer-reviewed journals or pre-prints to a consistent high standard.
- Available in standard formats [MITAB](#) or [PSI-MI XML 2.5](#).
- Provided by a network of participating major public domain databases.

List of IMEx members

- [DIP \(Active\)](#)
- [HPIDB \(Active\)](#)
- [IntAct \(Active\)](#)
- [MBInfo \(Active\)](#)
- [MINT \(Active\)](#)
- [MatrixDB \(Active\)](#)
- [Molecular Connections \(Active\)](#)
- [I2D \(Active\)](#)
- [InnateDB \(Active\)](#)
- [UCL-BHF group, UCL London \(Active\)](#)
- [UniProt group \(Active\)](#)
- [Swiss-Prot group, SIB \(Active\)](#)
- [EMBL-EBI \(Active\)](#)
- [BioGRID \(Observer\)](#)
- [PrimesDB \(Observer\)](#)
- [MPact \(I\)](#)
- [BIND \(In](#)
- [MPIDB \(I](#)

Available
Interaction
Network Data

<http://www.imexconsortium.org/about/>

Integrated Data Sources

Open Source

- Pathway Commons
- BioGRID
- MiMI (Michigan Molecular Interactions)
- STRING (Search Tool for Retrieval of Interacting Genes/Proteins)
- Genes2Network
- VisANT (Integrative Visual Analysis Tool)
- BIOBASE

Proprietary

- IPA (Ingenuity Pathway Analysis)
- MetaCore



Integrative Visual Analysis Tool for Biological Networks and Pathways

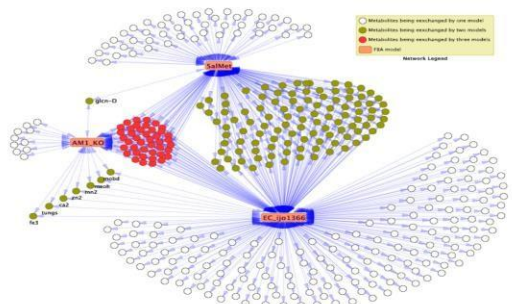
Home News Features Samples VisANT Community Documents Run & Download Database Help

VisANT 5.0

Visual analyses of metabolic networks in cells and ecosystems

V5 Manual

Download




Michigan Molecular Interactions

- MiMI:** Formerly called the Michigan Protein Database (MIPD). **It is currently available [here](#).** MiMI is an ongoing project supported by [NCIBI](#). Collaborators include the UM Medical School. Additional sponsors include the [NSF](#), [NIH](#), [MCBI](#), and [Microsoft](#).
- Biological Texts:** There is a great deal of interest on understanding biological texts automatically. There are also a large number of biological databases dealing with a variety of topics. The goal of our project is to be able to combine these two efforts by storing the information extracted from the text parsing into an XML database with a proper data model. With all the information managed by a database engine, complex queries can be issued and data mining can be applied.
- Integrated Protein Information:** Protein data, from sequence and structure to interaction, is being generated through many diverse methodologies; it is stored and reported in numerous forms and multiple places. The magnitude of the data limits researchers' abilities to utilize all information generated. Effective integration of protein data can be accomplished through better data modeling. Using an appropriate data model can enable researchers to exploit data more fully. Using [TIMBER](#) as our repository, we are developing a data model which allows fuller usage of the current protein data.

BioGRID 4.4

home help wiki projects tools contribute stats downloads partners about us

Welcome to our Database of Protein, Genetic and Chemical Interactions

BioGRID is a biomedical interaction repository with data compiled through comprehensive curation efforts. Our current index is version **4.4.216** and searches **81,370** publications for **2,570,689** protein and genetic interactions, **29,417** chemical interactions and **1,128,339** post translational modifications from major model organism species. All data are **freely** provided via our search index and available for download in many standardized formats.

BioGRID Statistics | Latest Downloads

Q Search BioGRID: By Protein/Gene

Search by Protein/Gene Identifiers ...

All Organisms

Submit Identifier Search

Advanced Search Helpful Search Tips Featured Datasets

BioGRID COVID-19 Coronavirus Curation Project

Search BioGRID for SARS-CoV-2 Protein Interactions | Download SARS-CoV-2 and Coronavirus-Related Interactions

Related Resources

BioGRID ORCS - An open repository of CRISPR screens

The BioGRID Open Repository of CRISPR Screens (ORCS) is a publicly accessible database of CRISPR screens compiled through comprehensive curation of all genome-wide CRISPR screen data reported in the biomedical literature. ORCS is updated on a quarterly basis and is fully searchable by

Partners

NIH ORIP OFFICE OF RESEARCH INFRASTRUCTURE PROGRAMS

CIHR IRSC

Singi



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Database: ESBIOBASE

ESBIOBASE (Elsevier BIOBASE) is a bibliographic current awareness database providing comprehensive coverage of the entire spectrum of biological research worldwide. Citations and abstracts are available within 15 days of receipt of the source material.

Each document is indexed using a unique versatile classification scheme that is updated annually to adapt to changing research trends. Records contain bibliographic and indexing information, including species indexing, an abstract, and author keywords, when available.

Citations and abstracts are available online within 15 days of the receipt of the source material.

Elsevier BIOBASE is not only an electronic equivalent of *Current Awareness in Biological Sciences (CABS)*, a well-established print product, but it contains some 100,000 additional items per year.

File Data

- 1994 to the present
- More than 7.2 million records (5/17)
- Updated weekly

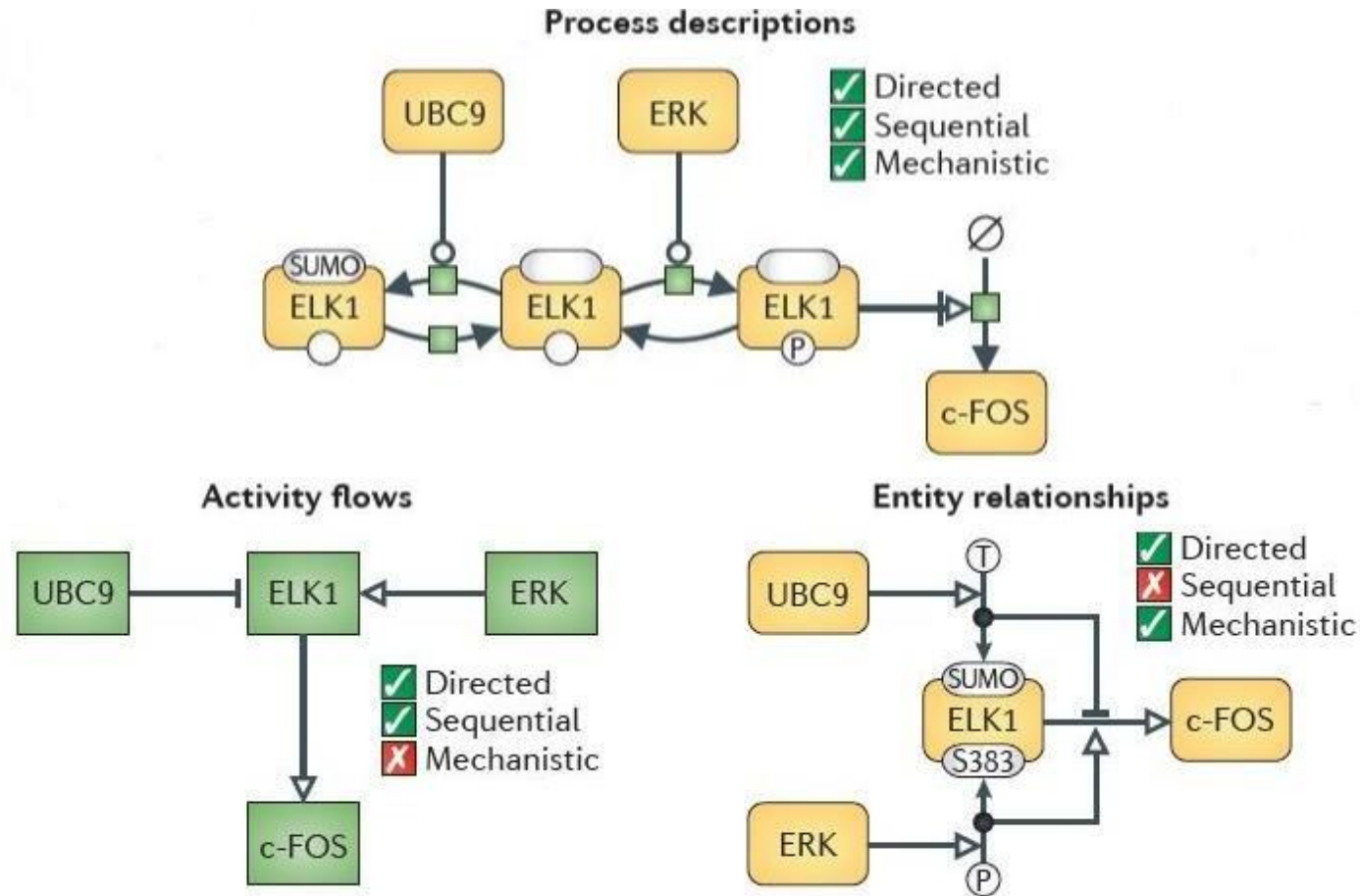
Subject Coverage

USING FORMALIZED DIAGRAMS TO REPRESENT Biological Networks



Systems Biology Graphical Notation (SBGN)

Describing mechanisms in a systematic fashion

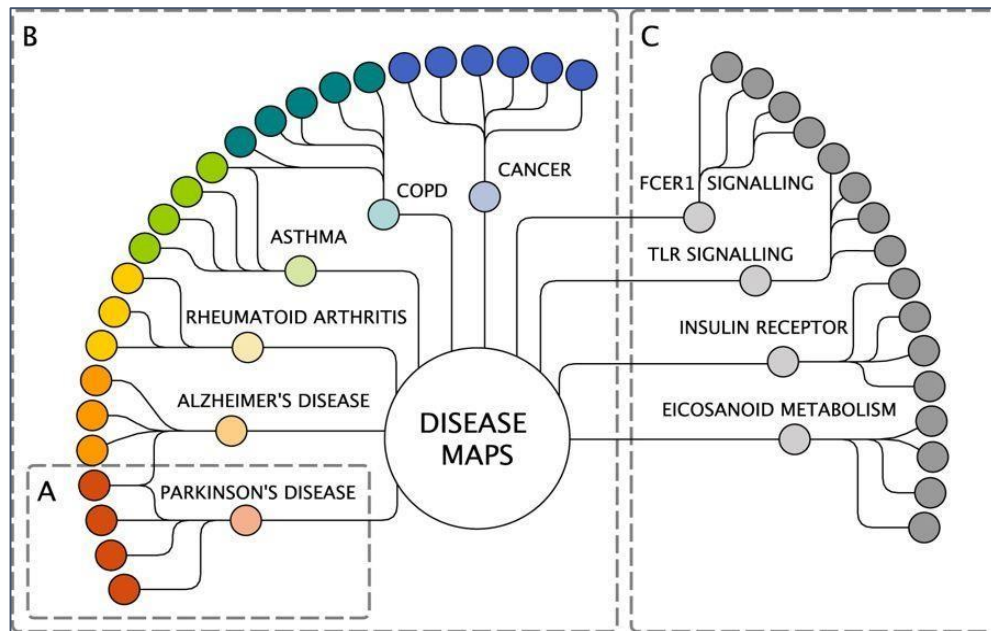


Process description diagrams of biological mechanisms

- Pioneering works of Prof. Hiroaki Kitano.
- Process description representation of signalling networks.
- First comprehensive disease-relevant extensive reconstructions of signalling pathways.
- Cancer Signalling Atlas – Curie Institute - <https://acsn.curie.fr/ACSN2/ACSN2.html>
- Disease Maps project <https://disease-maps.org/>
- REACTOME – PD like diagrams <https://reactome.org/>

The Disease Maps project

Large-scale open community effort



<http://disease-maps.org>

Key goals

Create

formalized mechanistic networks at different levels to investigate cellular machinery and host pathogen interactions.

Interpret

large-scale data sets and extract true information to understand how the disease takes place and progresses.

Develop

computational techniques, which can integrate large and heterogeneous data sets and combine them with prior knowledge.

Build

computational models that can provide insights into the mechanisms of interest.

Molecular interaction maps



Representations of biological processes (disease mechanisms) that are both human and machine-readable.



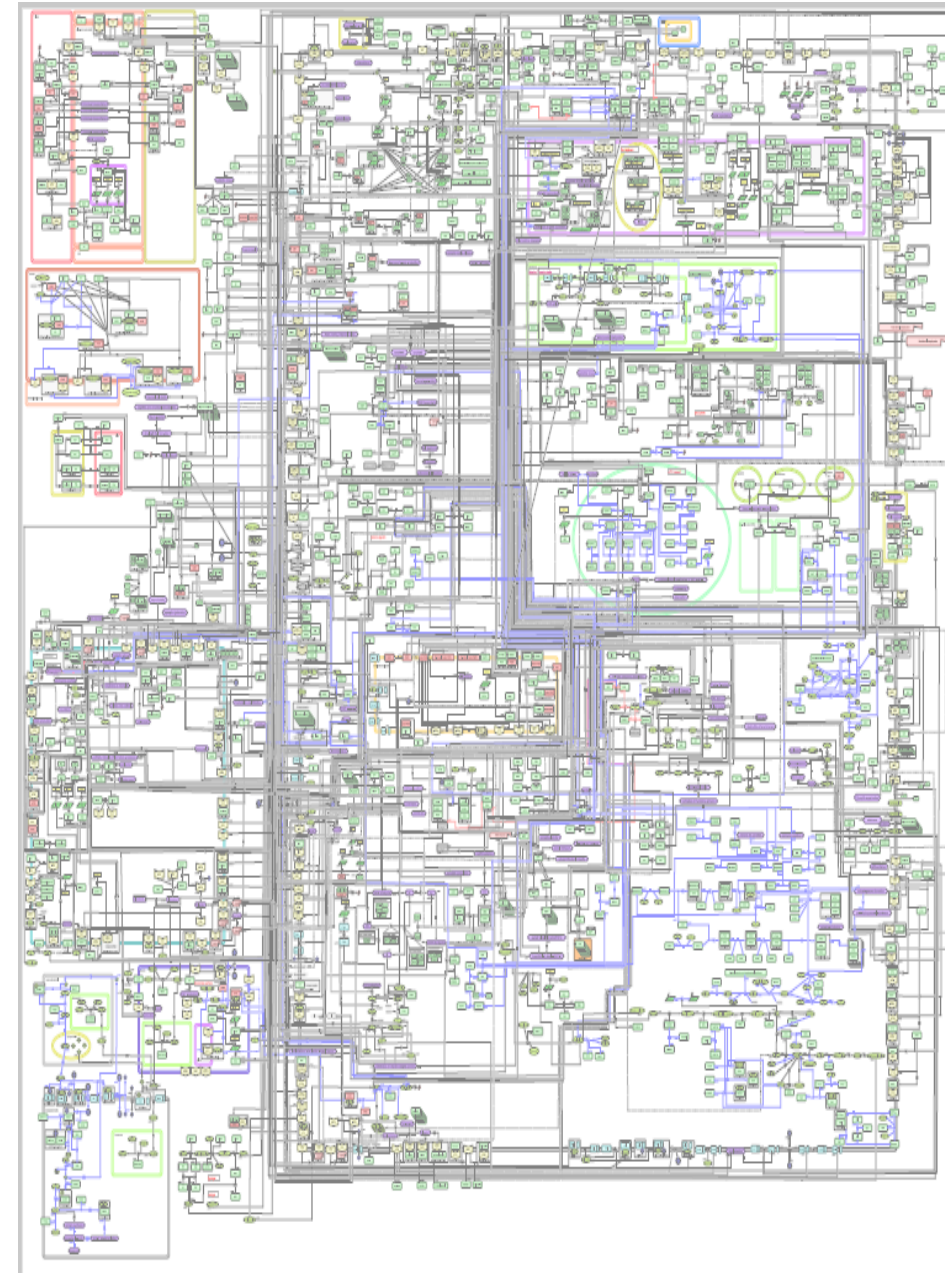
High quality source of knowledge (signalling pathways, gene expression, cellular phenotypes)—template for data visualization.



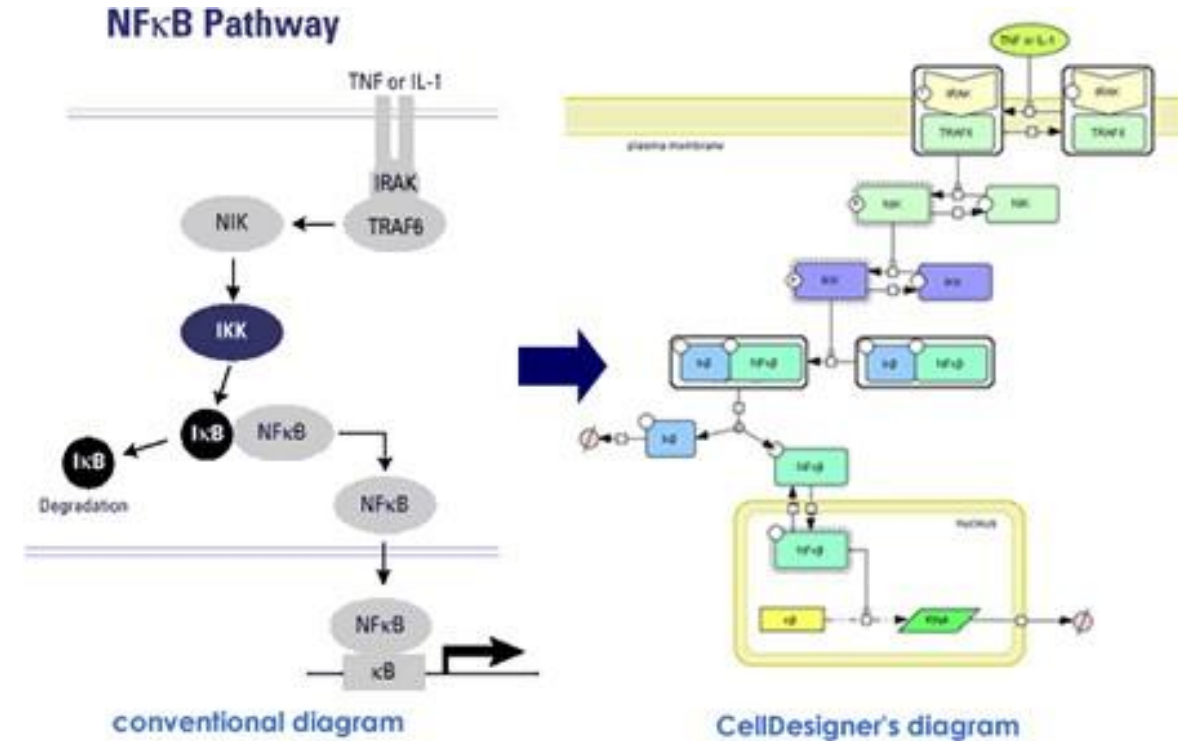
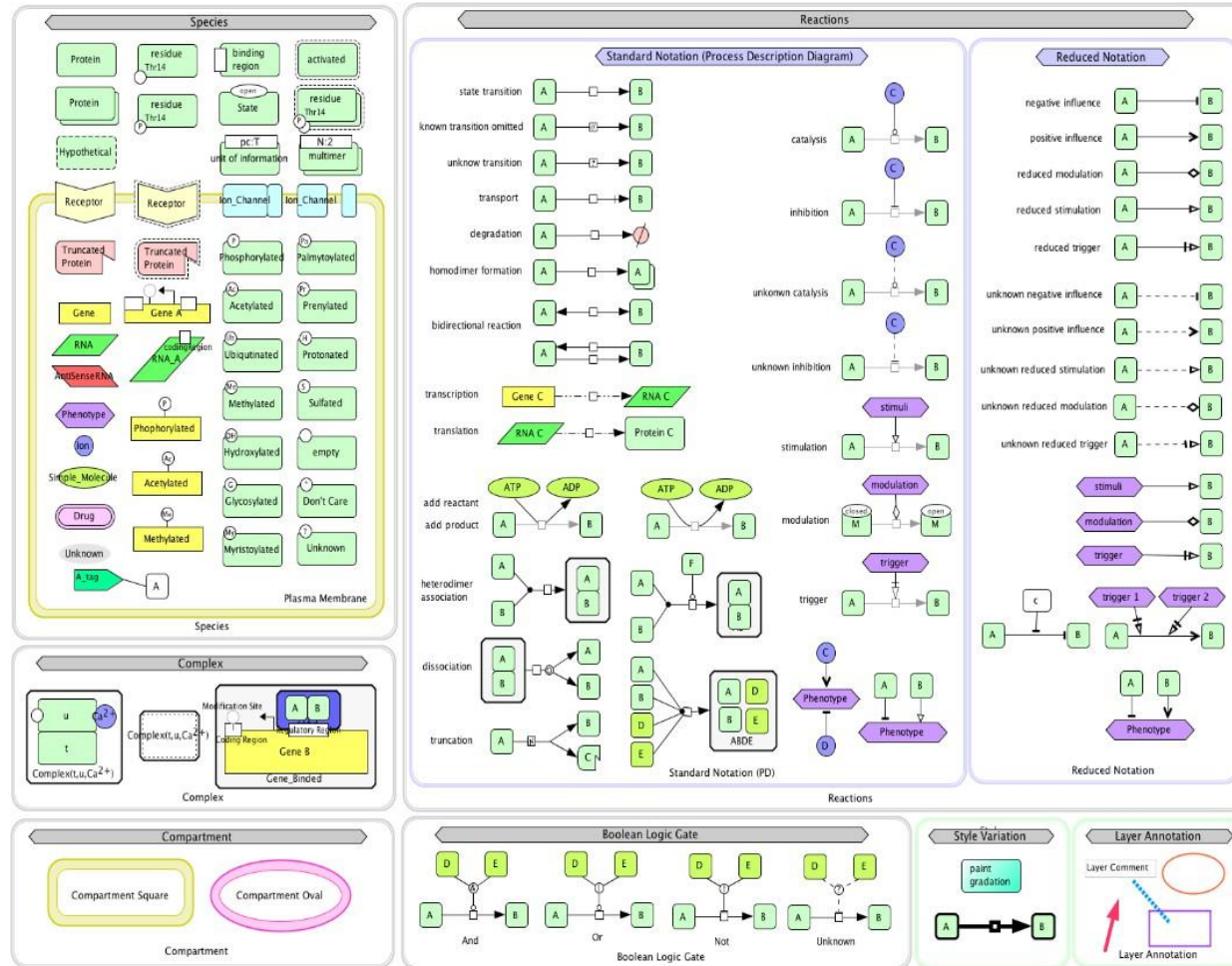
Can be seen and analyzed as a **complex network** (topology/structure).

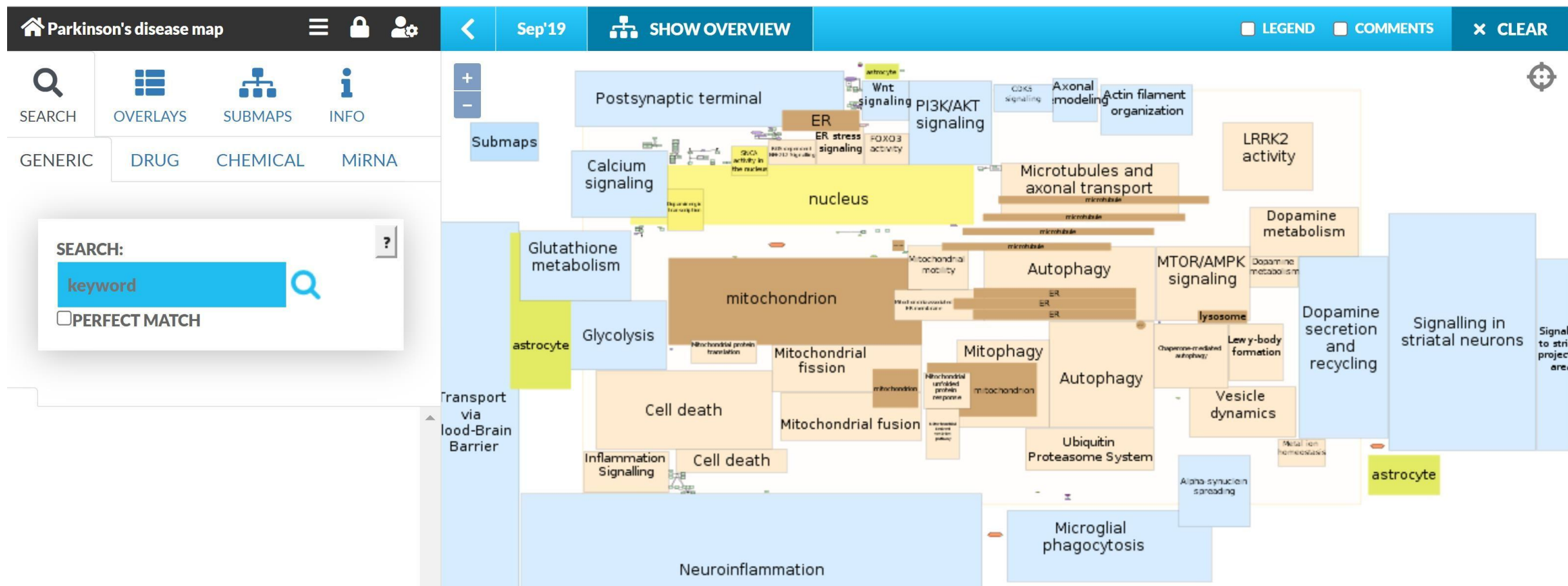


Can serve as a scaffold for a **mathematical model**.



Building molecular maps using CellDesigner



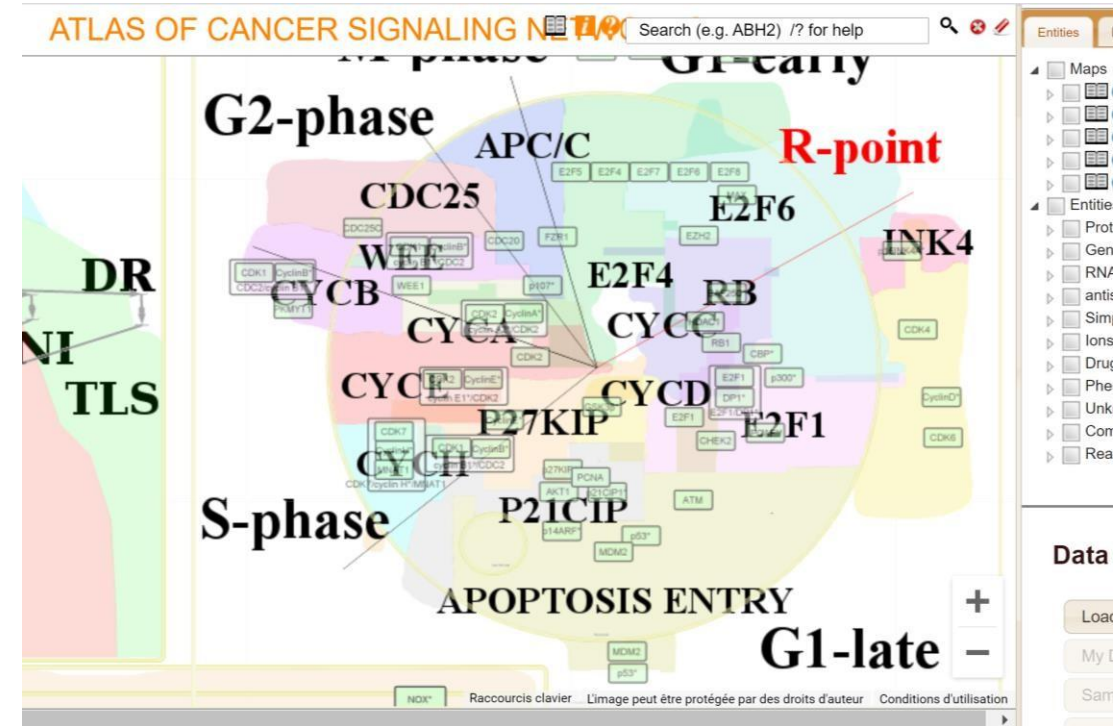
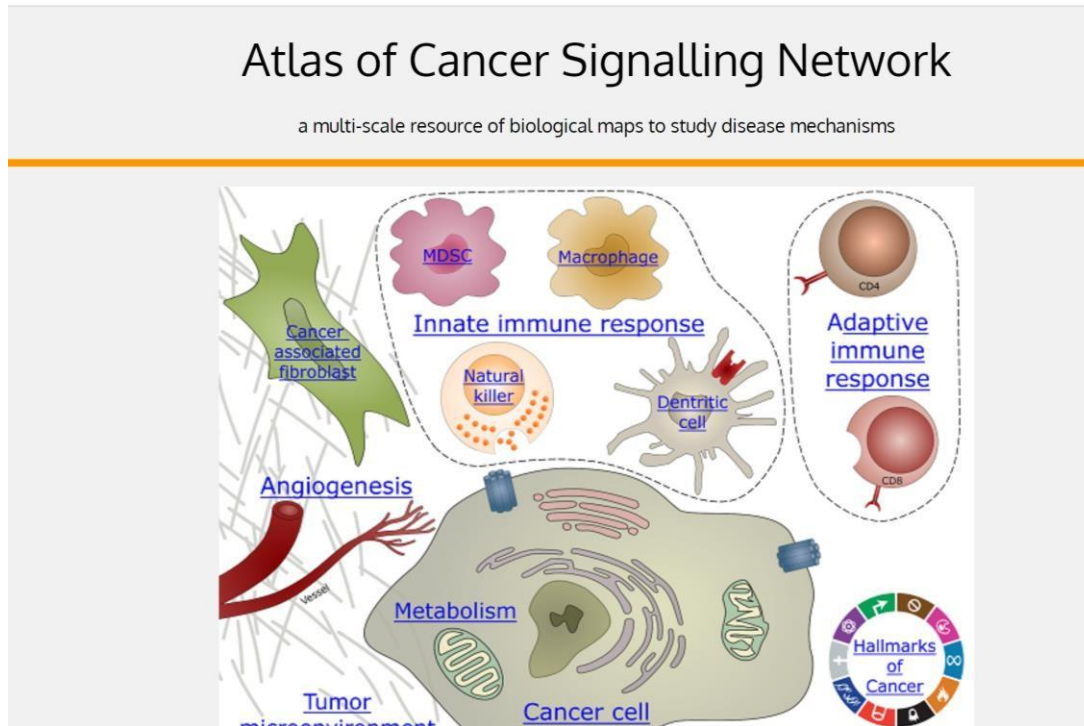


The Parkinson's map
<https://pdmap.uni.lu/Minerva/>

- Minerva platform
- Large scale curation effort
- Interdisciplinarity
- PD and AF

Atlas of Cancer Signalling Network

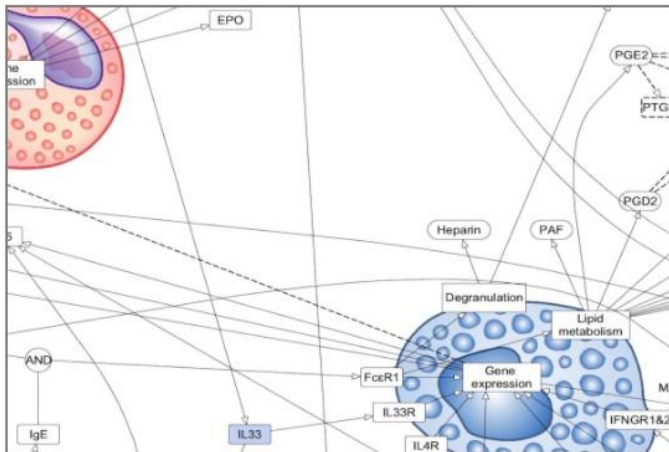
<https://acsn.curie.fr/navicell/maps/acsn/master/index.html>



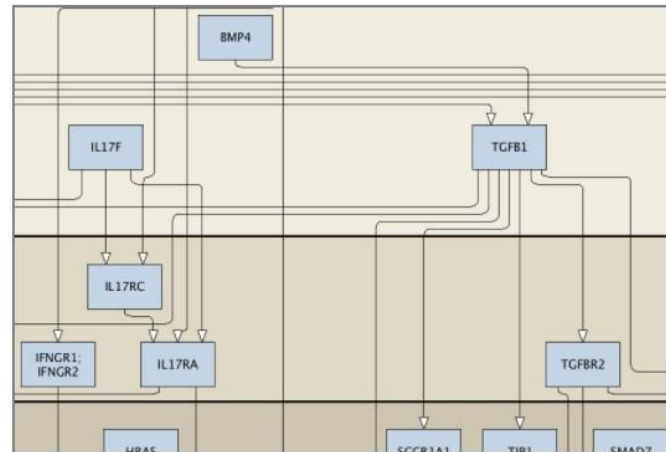
AsthmaMap architecture

The AsthmaMap includes three interconnected layers of granularity: Cellular Interactions (CI), a high-level overview; Molecular Relations (MR), the intermediate level of details; Biochemical Mechanisms (BM), the most detailed layer.

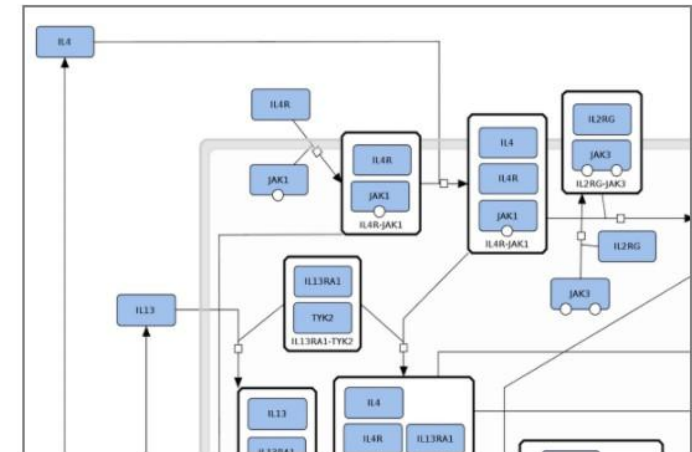
AsthmaMap Cellular Interactions



AsthmaMap Molecular Relations



AsthmaMap Biochemical Mechanisms



The Asthma Map

<https://asthma-map.org/>

- Granularity
- Different layers
- PD
- AF

The RA map

<https://ramap.uni.lu/minerva/>

- SBGN compliant, fully PD
- Detailed annotations using MIRIAM identifiers
- Executable through a map-to-model framework (PD→AF)

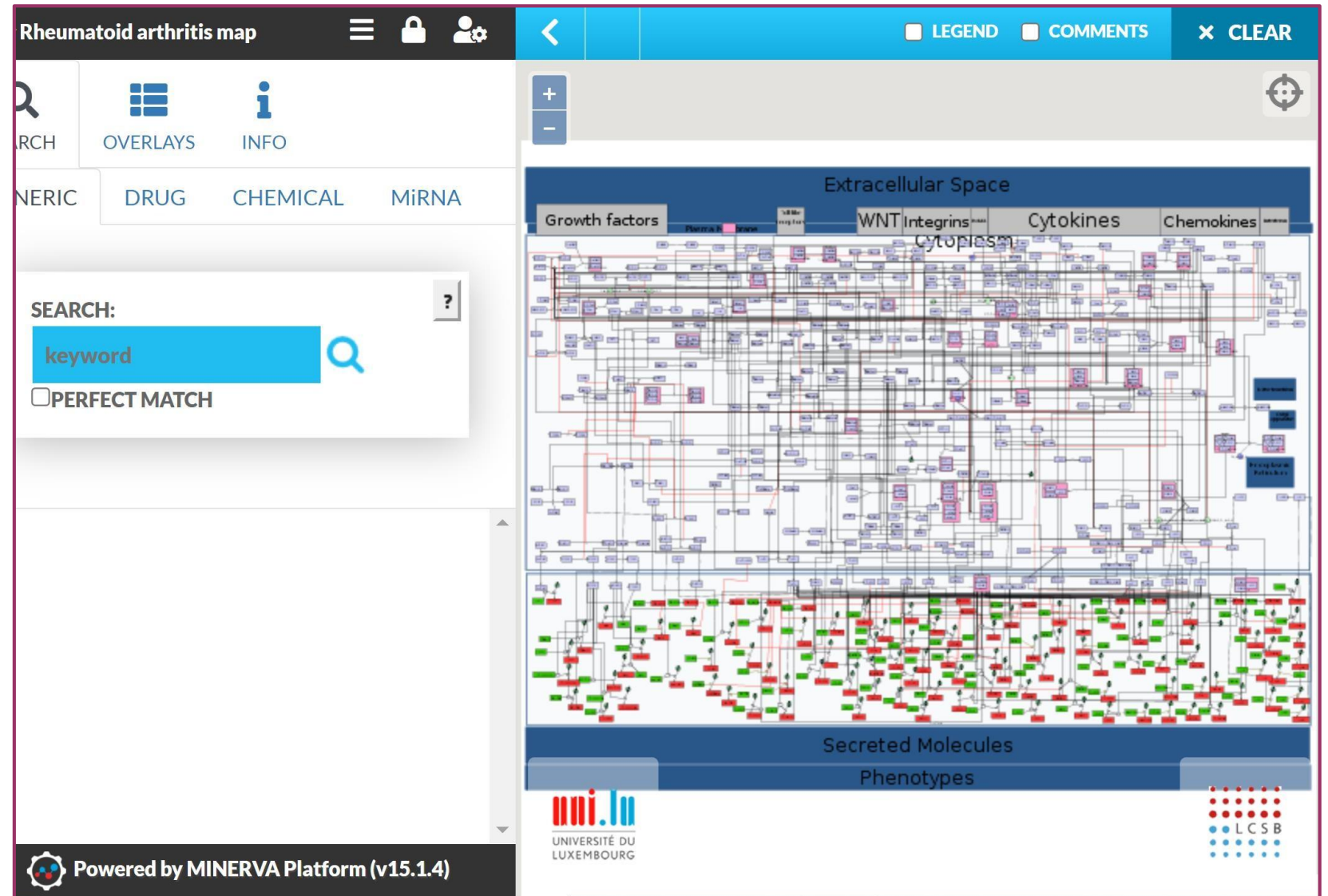
JOURNAL ARTICLE

RA-map: building a state-of-the-art interactive knowledge base for rheumatoid arthritis

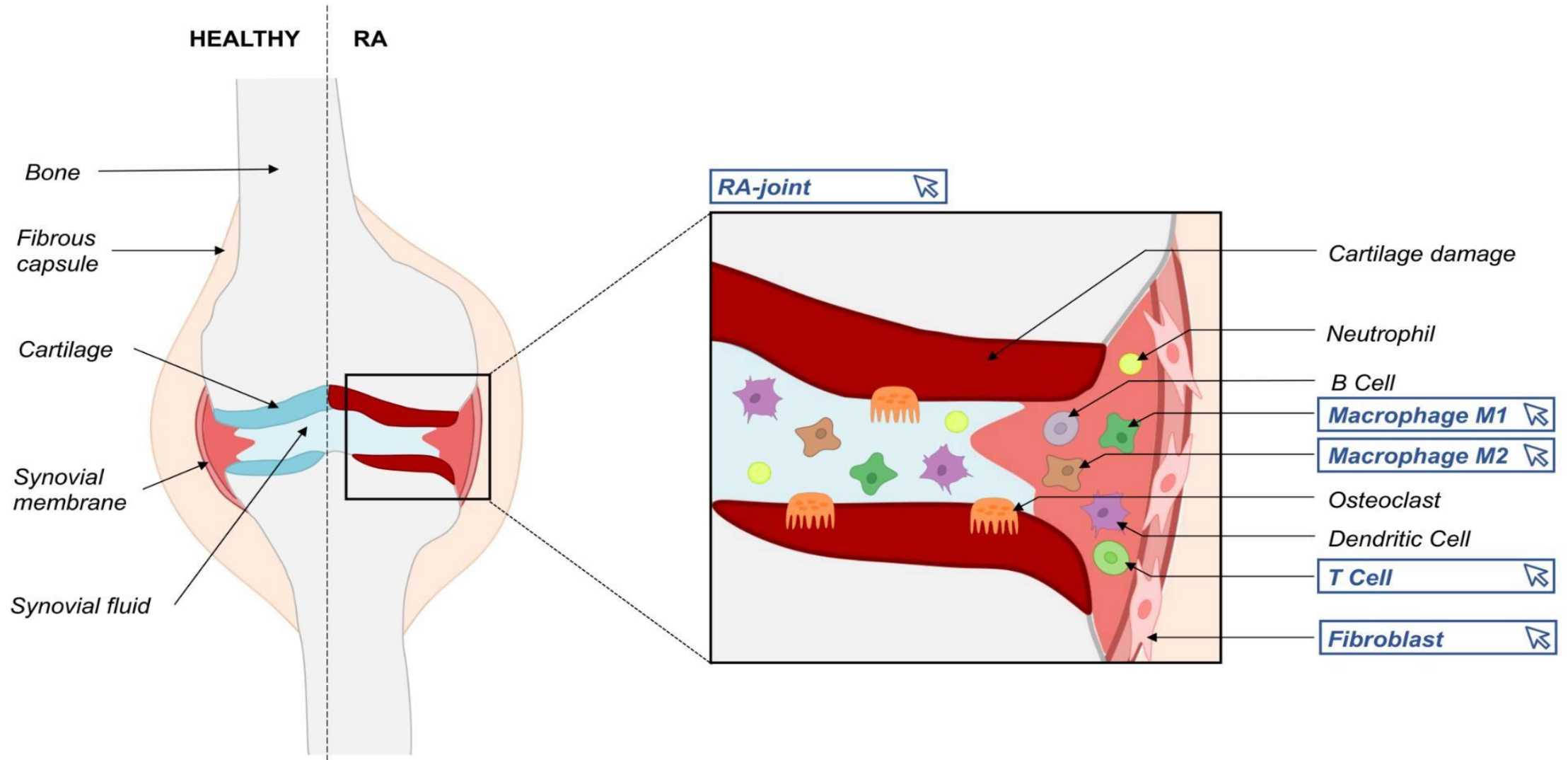
Vidisha Singh, George D Kalliolias, Marek Ostaszewski, Maëva Veyssiere, Eleftherios Pilalis, Piotr Gawron, Alexander Mazein, Eric Bonnet, Elisabeth Petit-Teixeira, Anna Niarakis

Database, Volume 2020, 2020, baaa017, <https://doi.org/10.1093/database/baaa017>

Published: 20 April 2020 Article history

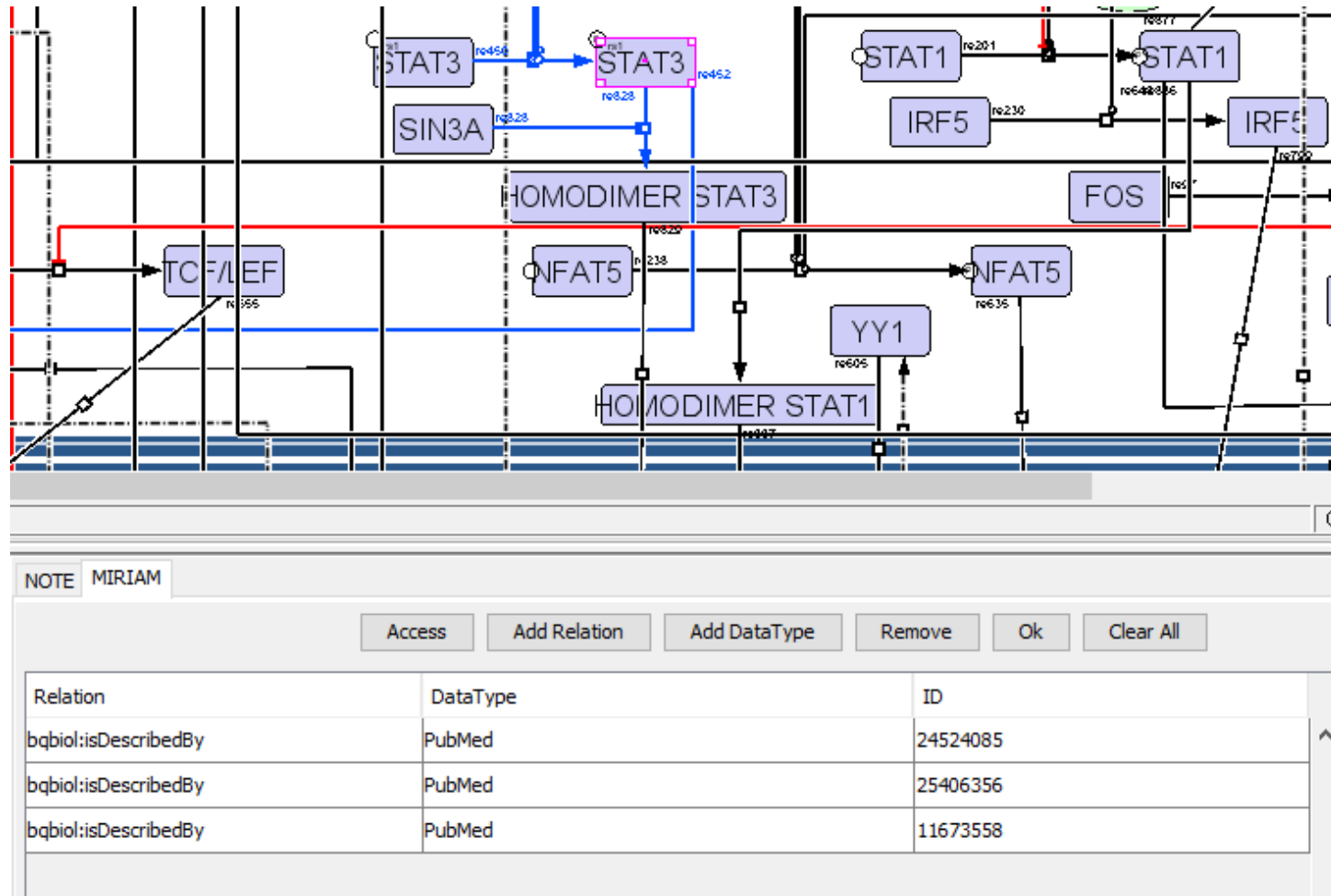


The RA –Atlas



Building molecular maps using CellDesigner

MIRIAM: Minimal Information Required In the Annotation of Models



- Facilitating interoperability and model reusability
- Annotations retrieved in the resulting model

Interactive knowledge base for RA using MINERVA –(Molecular Interaction NEtwoRk VisuAlization) platform

SEARCHOVERLAYSINFO

GENERICDRUGCHEMICALMIRNA

SEARCH:

il6

PERFECT MATCH

1

RNA: IL6

+

Compartment: Nucleus

Full name: interleukin 6

Symbol: IL6

Former symbols: IFNB2

Synonyms: BSF2, HGF, HSF, IL-6

Annotations:

Source: Annotated by curator

[1] PUBMED(10688908)

[2] PUBMED(17652167)

[3] PUBMED(2462501)

[4] PUBMED(18281366)

[5] PUBMED(18454843)

[6] PUBMED(24524085)

[7] PUBMED(22870451)

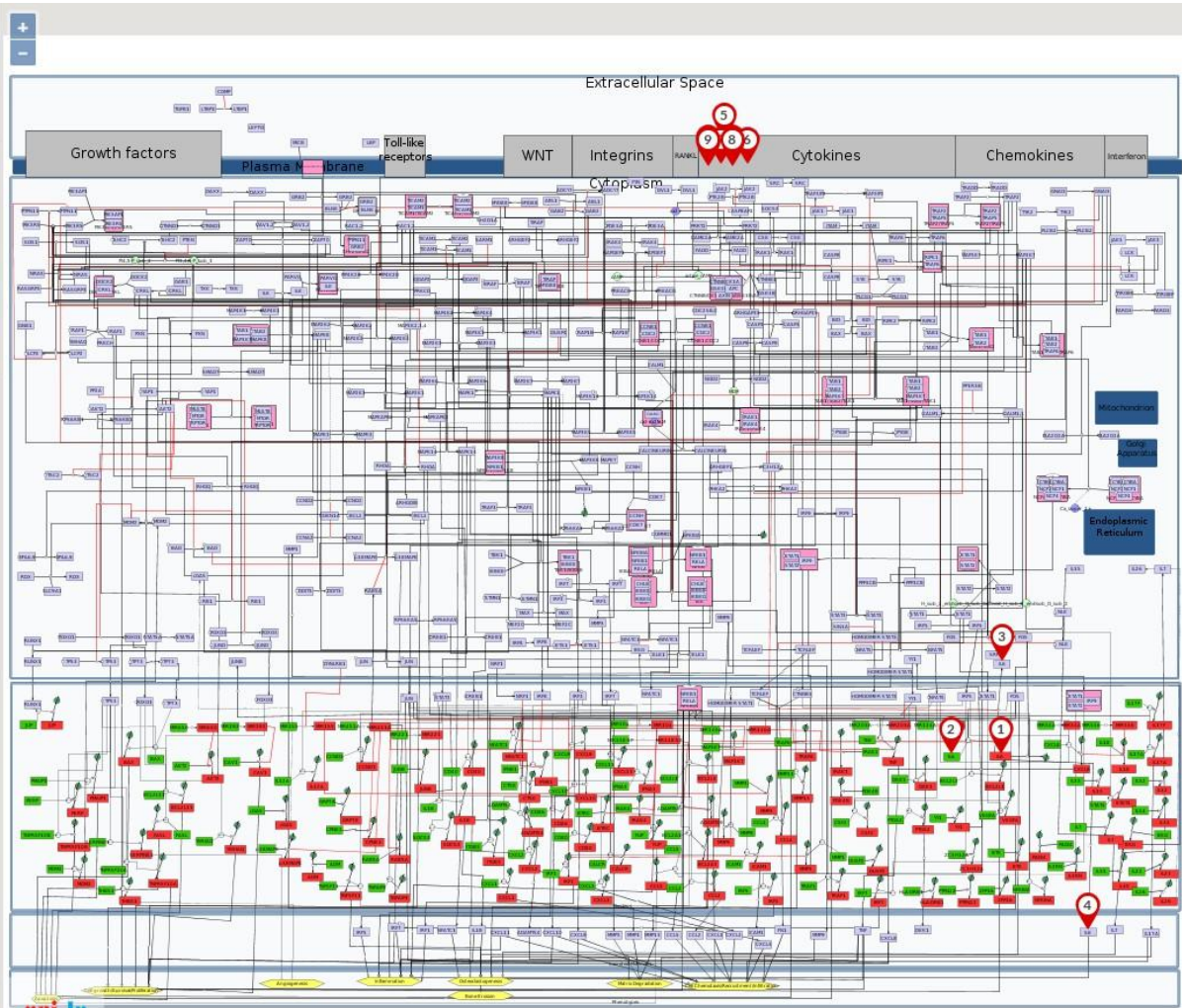
[8] PUBMED(8484679)

[9] PUBMED(12905466)

[10] PUBMED(28494214)

[11] PUBMED(18205922)

Powered by MINERVA Platform



Protein: IL6

Interacting drugs

Name	References
Search for available targets	

Interacting chemicals

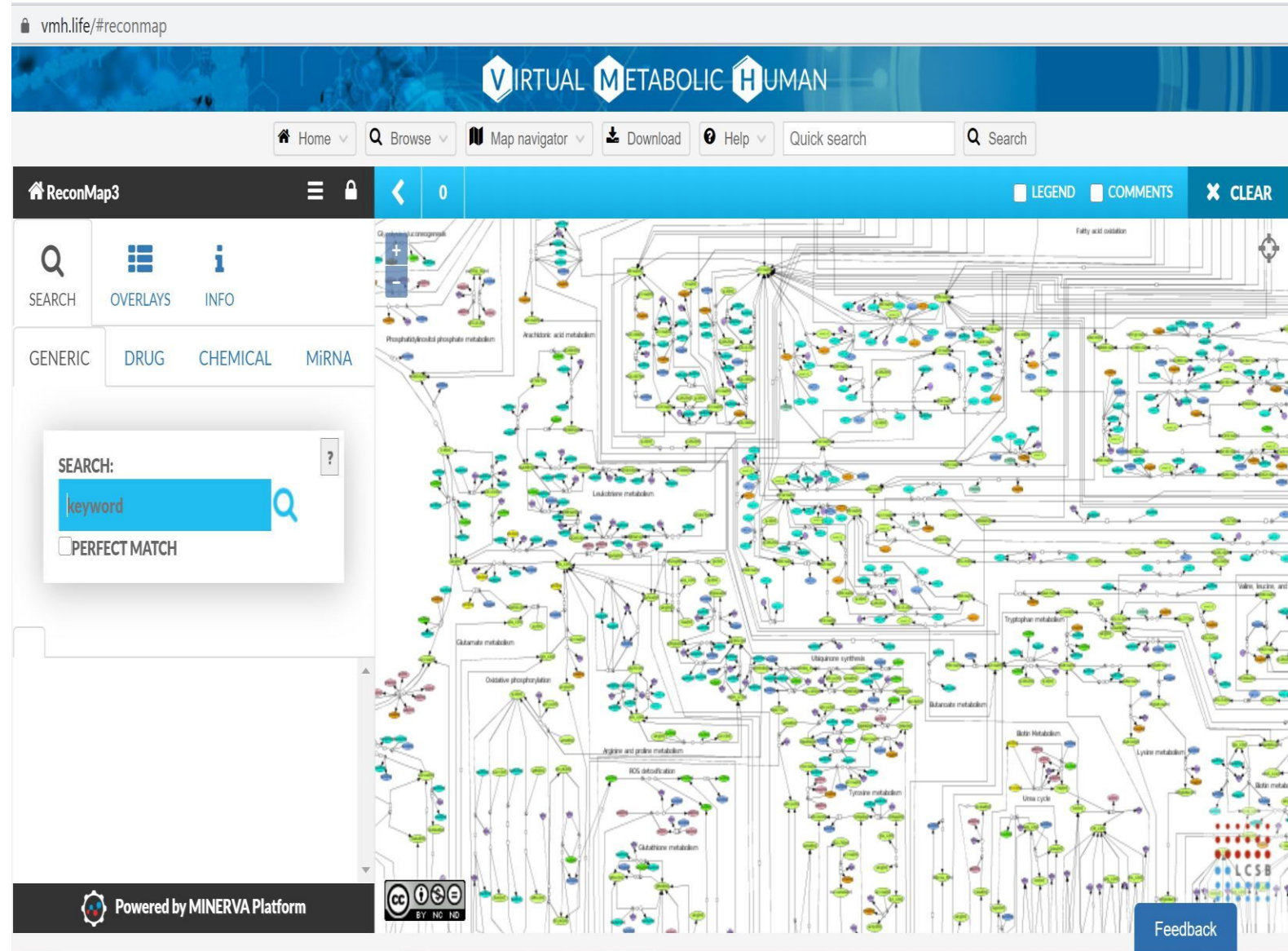
Name	References
Search for available targets	

Interacting Micro RNAs

Name	References
Search for available targets	

The ReconMap

- ReconMap content obtained from the Virtual Metabolic Human database (VMH, <http://vmh.uni.lu>).
- Recon-derived simulation results can be visualized on ReconMap using a new extension to the COBRA Toolbox (Schellenberger et al., 2011).
- User can perform a simulation, e.g. Flux Balance Analysis, using the COBRA toolbox function 'optimizeCBmodel', then call the function 'buildFluxDistLayout' to write the input file for a context-specific ReconMap Overlay.



Menu

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- [Model Repository](#)
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Related Link

[systems-biology.org](#)

[PANTHER database](#)
[BioModels.Net](#)

The Systems Biology
Institute
Kitano Symbiotic
Systems Project



CellDesigner™: A modeling tool of biochemical networks



Current Release Version: CellDesigner 4.4.2

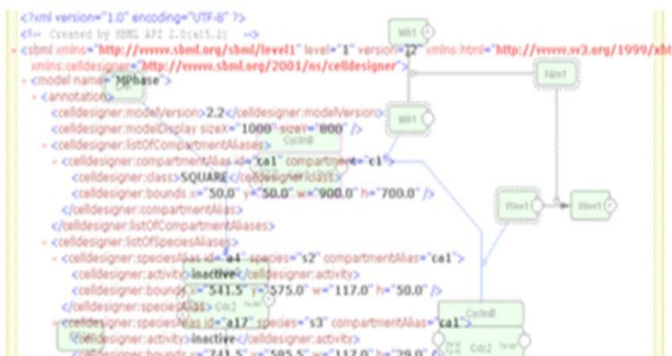
macOS Catalina and Ubuntu 18.04 support + Plugin APIs enhances + BioModels new API support + Garuda enabled + bug fixes. find out more...

You do not have to install JVM separately as it is included in the installer.

Check also:

- CellDesigner on [Garuda platform](#) (Ver4.4.1 Win / Mac)
- Plugins / Utilities
 - CellDesigner Plugin API Document (ver4.4.1)
- Models built with CellDesigner
- BioModels.net models simulation results with CellDesigner 4.0

What is CellDesigner™



Headlines

[CellDesigner 4.4.2](#) Mac installer updated for Catalina Support (2020/3/30)

[CellDesigner 4.4.2](#) is now available (2019/05/20)

[CellDesigner on Garuda platform](#) (2017/2/14) Garuda enabled Ver4.4 is available as Ver4.4.1 [Win](#) / [Mac](#).

[PhysioDesigner 1.0](#) is available. PhysioDesigner can embed CellDesigner's SBML model into its multilevel physiological PHML model. (2014/08/01)

[CellDesigner 4.4](#) is now available (2014/07/12)

Advertisement



Systems Biology
and Applications

An open access, multi- and cross-disciplinary journal dedicated to publishing the finest research on systems biology and all its approaches.

In partnership with  The Systems Biology Institute



yworks
the diagramming experts

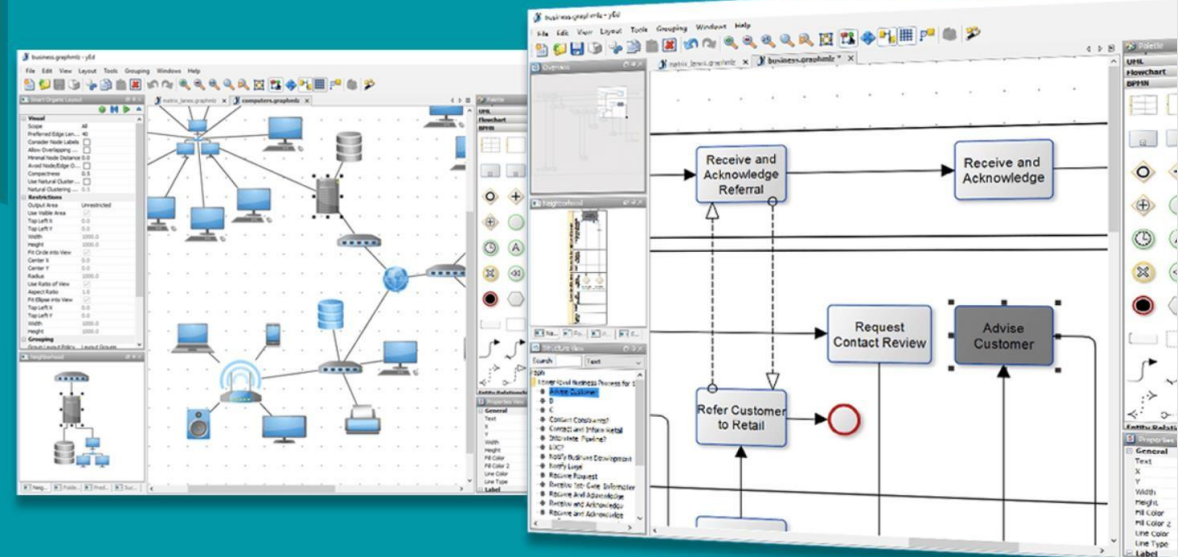
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yEd Graph Editor: High-quality diagrams made easy

DOWNLOAD NOW

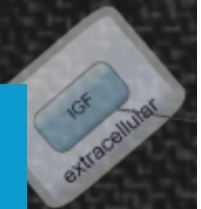
New Online
Version

yEd Live



newt Pathways Simplified

View, design, and analyze pathways in SBGN and more...

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Welcome to Newt Pathway Viewer & Editor

Newt is a free, web based, open source viewer and editor for pathways in **Systems Biological Graphical Notation (SBGN)** and **Simple Interaction Format (SIF)**. It was written with a series of libraries and extensions based on **Cytoscape.js** with utmost customization in mind.

Launch 

What distinguishes Newt from other viewers and editors for biological maps can be summarized as:

Search

ENHANCED BY



Latest News

Newt release

April 1, 2020

Newt 3.0 was released to include some **new features** such

... ..

Prerequisites for high-quality biochemical interaction maps:

- **Accurate** – correctly represents our empirical knowledge.
- **Reusable** – well annotated and referenced.
- **Comprehensive** – accounts for all known reactions within the selected scope.
- **Machine readable** – can be processed and analyzed using computers.

- **Executable** – corresponds to a computational model that can be simulated.
- **Functional** – can explain the known system-level behavior of the biological network.

(Inspired by *Systems Biology*, ed. Nielsen and Hofmann, Chapter 8, Wiley-VCH, 2016
and *Community-driven roadmap for integrated disease maps*, Ostaszewski et al., 2018)



Adding a dynamical layer

“Why build models?” Jay Bailey

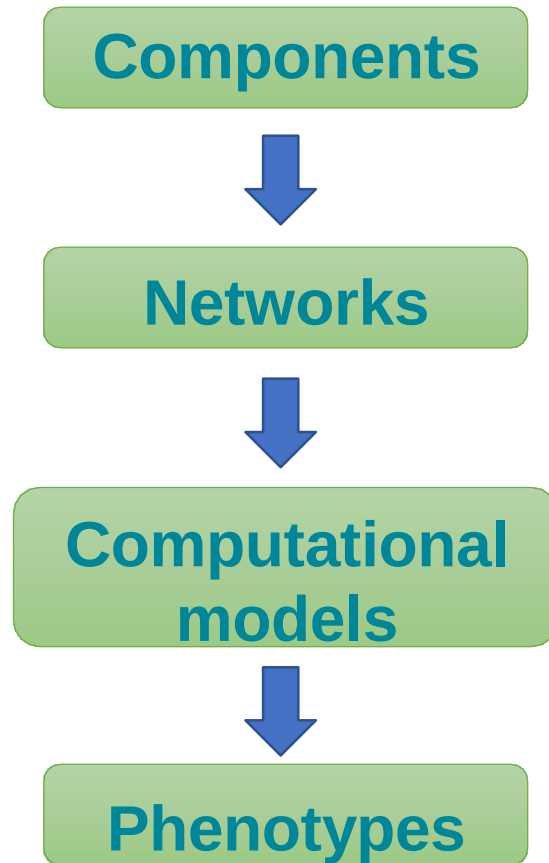
- To organize disparate information into a coherent whole.
- To think (and calculate) logically about what components and interactions are important in a complex system.
- To discover new strategies.
- To make important corrections to the conventional wisdom.
- To understand the essential qualitative features.



Computational models

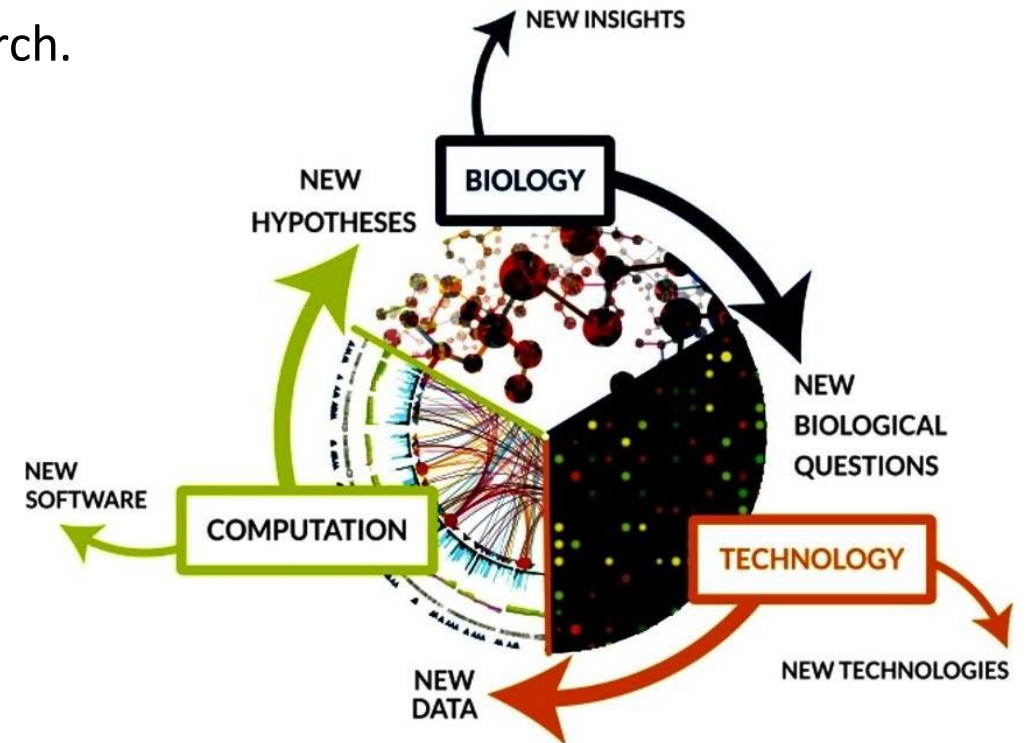
- Models are used to represent actual quantitative/qualitative relations between the molecules in the system.
- Abstract representations of biological processes
- Have an inherent execution scheme attached to the model.

Central dogma of computational systems biology

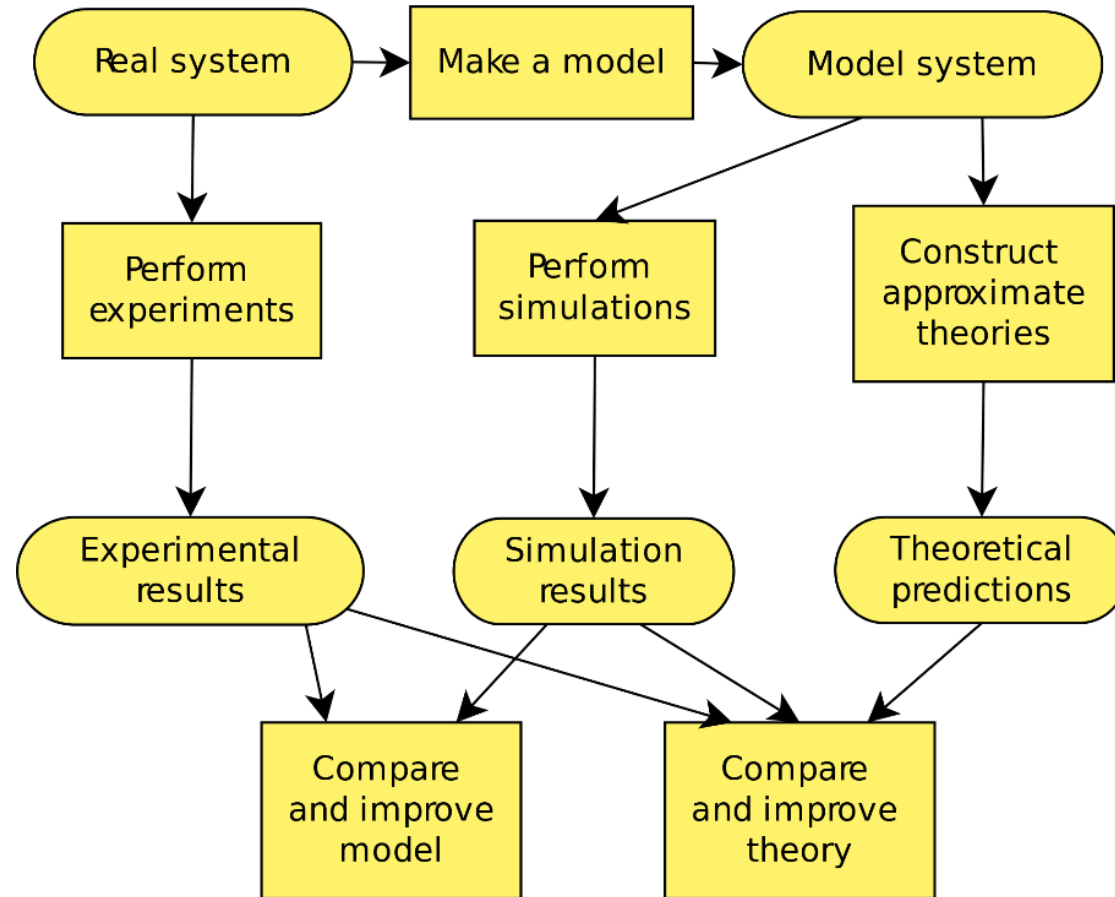


Model building: a step by step process

- Generation of the structural model based on a literature search.
- Compilation of a calibration dataset using experimentally validated biological knowledge.
- Model fitting to the experimental data.
- *In silico* simulations to generate predictions.
- Validation of predictions through experimental testing.
- Model refinement by feeding newly generated experimental data back to the model.
- Generation of novel hypotheses.
- Reiteration.

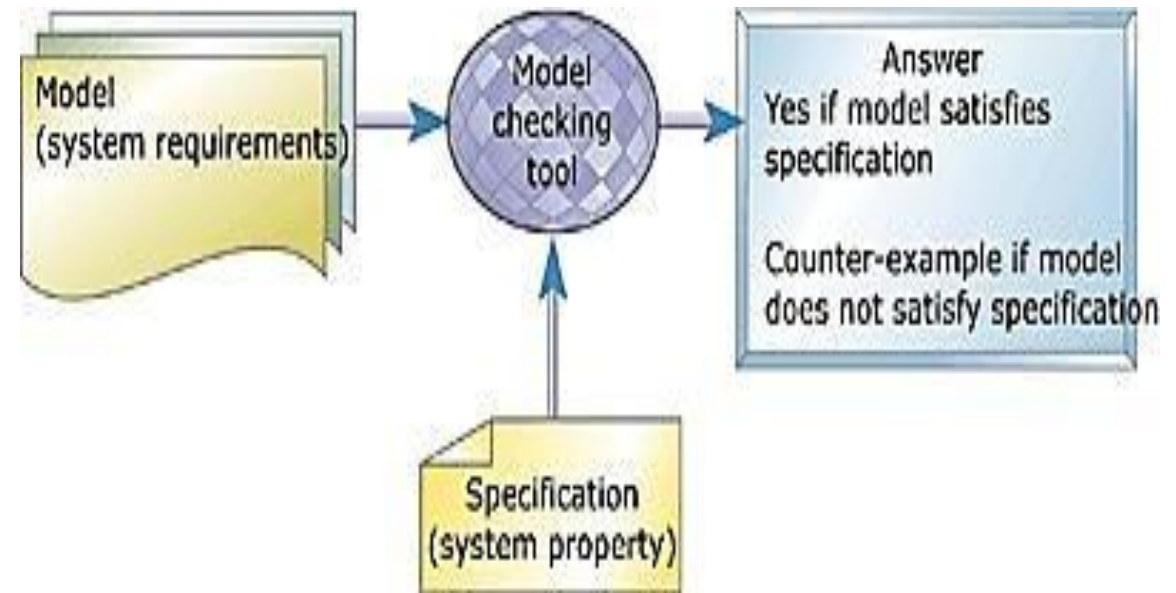


***“In theory, there is no difference between theory and practice.
But in practice, there is.” “The Yale Literary Magazine”, February 1882; Benjamin Brewster***



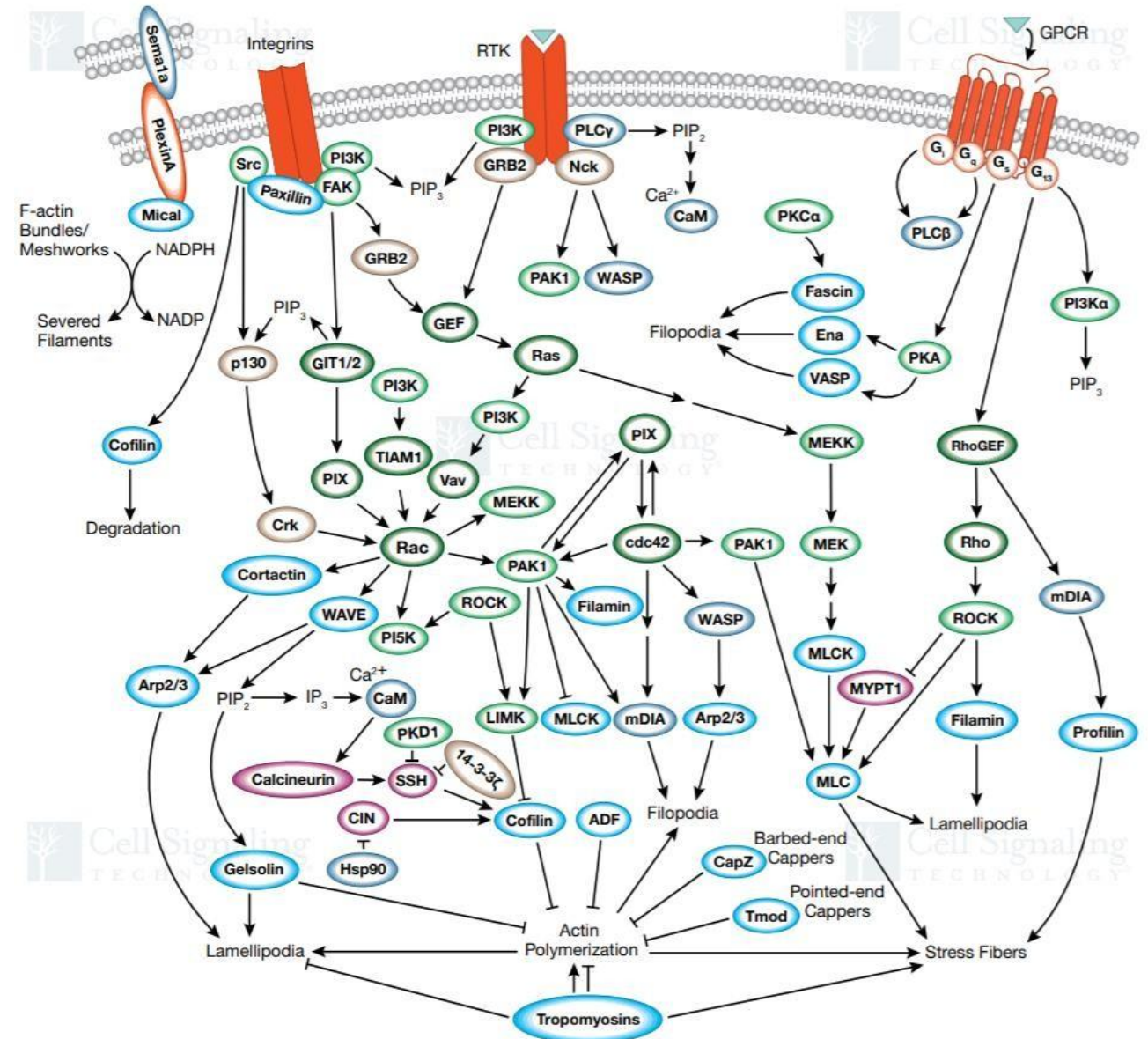
Model checking techniques - Comparing mechanistic models to specifications

- A formal model of the system under study is constructed.
- Experimental evidence is formalized as specifications (observations).
- Model checking is used to ensure that the model reproduces the experimental observations.
- **Mismatch with experimental observations:** model should be refined by additional information.
- **Match with experimental observations:** could lead to further querying and testing of the model to suggest further experimental studies.



Reachability analysis

- Extra-cellular molecules trigger a response inside the cell by initiating a signal at special membrane receptors.
- Signal is transmitted to reporters/targets through various chains of interactions among proteins.
- Understanding whether such a signal can reach from membrane receptors to reporters is essential in studying the cell response to extra-cellular events.



COMPUTATIONAL (BIO) MODELING METHODOLOGIES LEARNING METHODOLOGIES

Hypothesis-driven

(Mostly) Data-driven

PROCESS ALGEBRA
An example in Beta Binder

$$((\beta(x, \Delta_e)[E]) \parallel (\beta(y, \Delta_i)[I]))$$

↓ join operator

$$\beta^h(x, \Delta_e) \beta^h(y, \Delta_i)(E \parallel I)$$

$\beta(\)$ active binding site
 $\beta^h(\)$ hidden binding site
 $\parallel$ parallel operator
Enzyme (E) Inhibitor (I)

RULE-BASED SYSTEMS

EGFR(ECD!1,aa1092~Y).EGFR(ECD!1,aa1092~Y) $\xrightarrow{1, k_1}$ EGFR(ECD!1,aa1092~pY).EGFR(ECD!1,aa1092~Y)

(An example in BioNetGen)

PETRI NETS

Enzyme 1, Enzyme 2, Enzyme 3
Reaction A, Reaction B, Reaction C
Signal Transduction Network

PLACE
TRANSITION NOT ENABLED
TOKEN
TRANSITION ENABLED

SUPERVISED LEARNING

Clinical data: demographic, genetic, cognitive assessment data
MRI, CT, PET
Deep Learning
Feature extraction: $f_w(\text{data})$
Classifier
Prediction
Validate performance (acc, auc)

Advantages:
• Data transformations learnt during training

Issues:
• Typically many parameters $\rightarrow$ overfitting
• Difficult to map results back to informative features

LOGIC-BASED

$$rl(br_{t2.1}) \stackrel{\text{def}}{=} \forall t, n. T(t) \otimes C(n, \text{breast}, 1, [TGF\beta]) \rightarrow T(t + d_{11}(f)) \otimes A(n), f \in 0..2$$

FLUX BALANCE + OPTIMISATION

Vector of Fluxes
Metabolic network
Pareto front
Cellular target 2
Cellular target 1
Multicore

CONTINUOUS TIME AND HIDDEN MARKOV CHAINS

a) Alignment of protein sequences
b) Maximum likelihood
c) Set of amino acid substitution matrices
d) Structure predictions, prediction of evolutionary rates

UNSUPERVISED LEARNING

Variational autoencoder (VAE) probabilistic framework

Input, Latent representation, Output
Encoder, Decoder

Unsupervised neural models developed:
• DiffVAE
• Disentangled-DiffVAE
• Graph-DiffVAE

$$q_\phi(z|x) = \mathcal{N}(z; \mu, \text{diag}(\sigma^2))$$

$$p_\theta(x|z) = \prod_{i=1}^n p_\theta(x_i|z) = \prod_{i=1}^n x_i^{1+x_i} (1+x_i)^{1-x_i}$$

LATTICE-BASED

Concentration, Age

CHEMICAL MASTER EQUATION

System B: CdkA, pRb, EGFR, Wnt
CdkA200, CdkA200, CdkA200
Probability
Age of molecule

AGENT-BASED

Signaling Molecule (Communicating Agent)
Receptor (Reactive Agent)
Messengers (Cooperative Agents)
Nucleus (Reactive Agent)
Signal Transduction

BAYESIAN LEARNING

$$\ell(\beta_Z) = \arg \min_{\beta_Z} \left(\sum_{i=1}^n \delta_i \left\{ \mathbf{x}_{T,i}^T \beta_Z - \log \left[\sum_{j \in R(i)} \exp(\mathbf{x}_{T,i}^T \beta_Z) \right] \right\} + P_\lambda(\beta_Z) \right)$$

Network-based Cox Regression models

Network function Penalties: $P_{\lambda, \alpha}(\beta)$

ORNSTEIN-UHLENBECK
Branching Phylogeny

COMPLEX NETWORKS
Omic layer 1, Omic layer 2

GAME THEORY
Player 1, Player 2

	Altruism (+)	Spite (-)
Player 1	-	-
Player 2	+	+
	Cooperation (+)	Selfishness (-)

REINFORCEMENT LEARNING

State, Action, Reward, Environment, Agent

Interpretability/ non linearity/parameter estimation

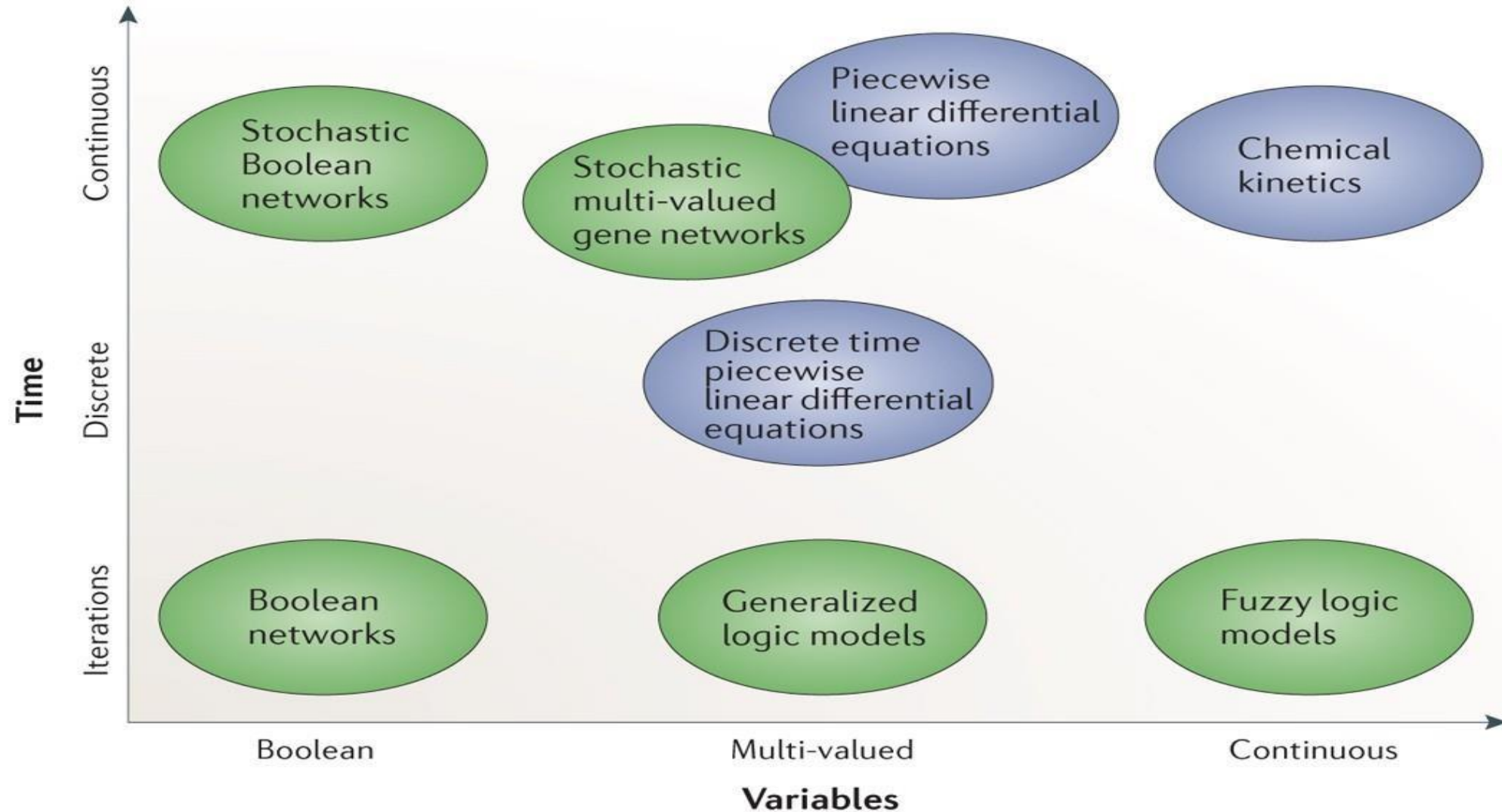
Quantitative versus qualitative models

Table 1

Comparison matrix of quantitative and qualitative models

	Quantitative model	Logic model
Suitable for	Time series	Phenotypes
Time representation	Linear representation	Abstract iterations
Variables	Quantitative	Qualitative
Mechanism representation	Yes	No
What can we do?	Compute concentrations and durations; evaluate the effect of parameter values	Compute state transitions and attractors (steady-states and cyclic attractors)
Data necessary to build the model	Molecular species, genes, interactions, biochemical processes	Activities, defined phenotypes, rules linking those
Data to parameterize and validate the model	Amount of molecular species, timecourses, quantitative phenotype	Perturbations of activities such as RNA interference, inhibitors, qualitative phenotypes
Advantages	Quantitative, precise; direct comparison with quantitative measurements; large existing toolkit	Easy to build; easy to compose; easy simulation of perturbations
Weaknesses	Requires quantitative knowledge of initial conditions and kinetics	Cannot provide quantitative predictions; difficult to choose between alternative behaviours

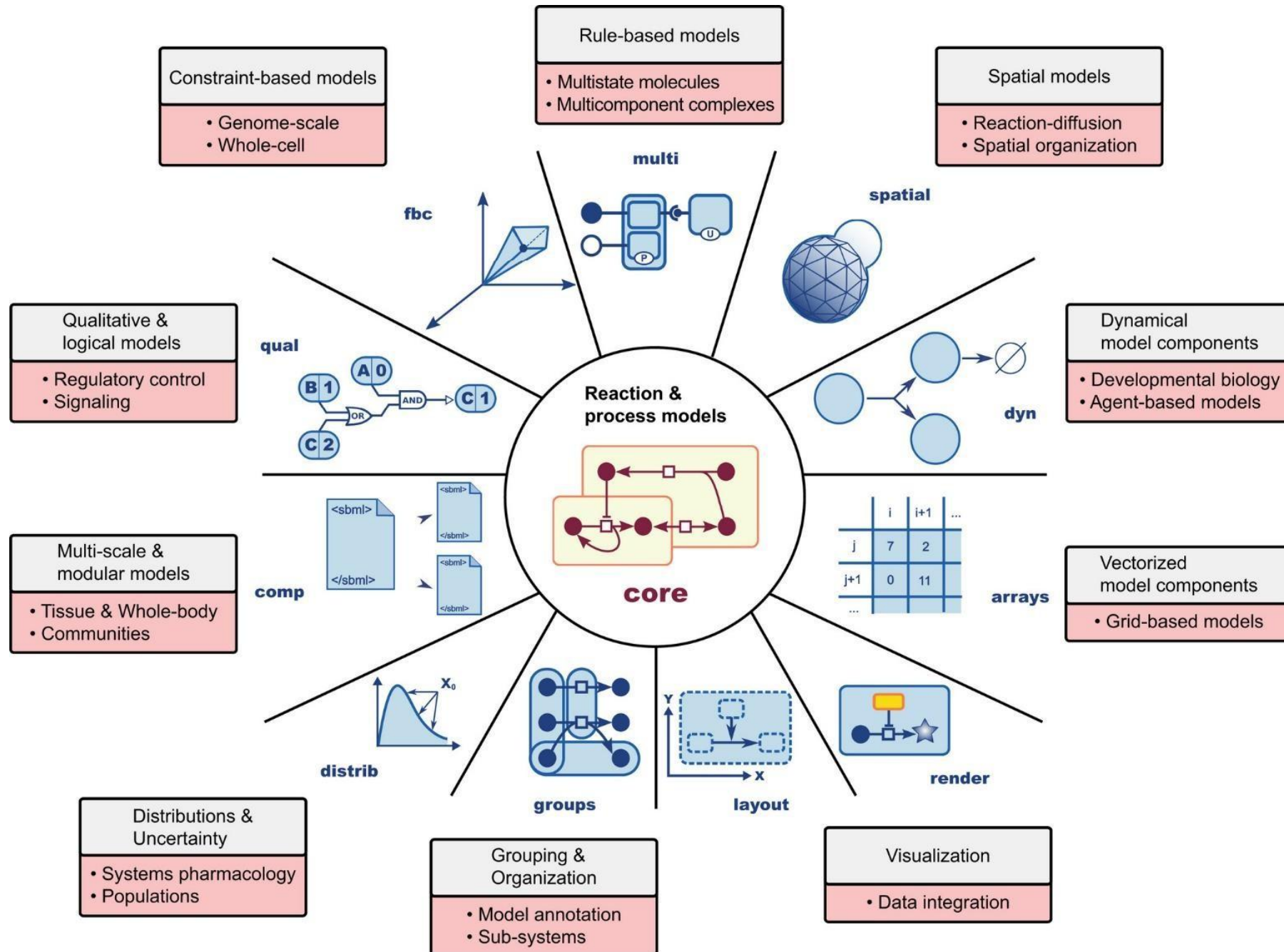
Granularity of time representation and variable values for various modelling approaches.



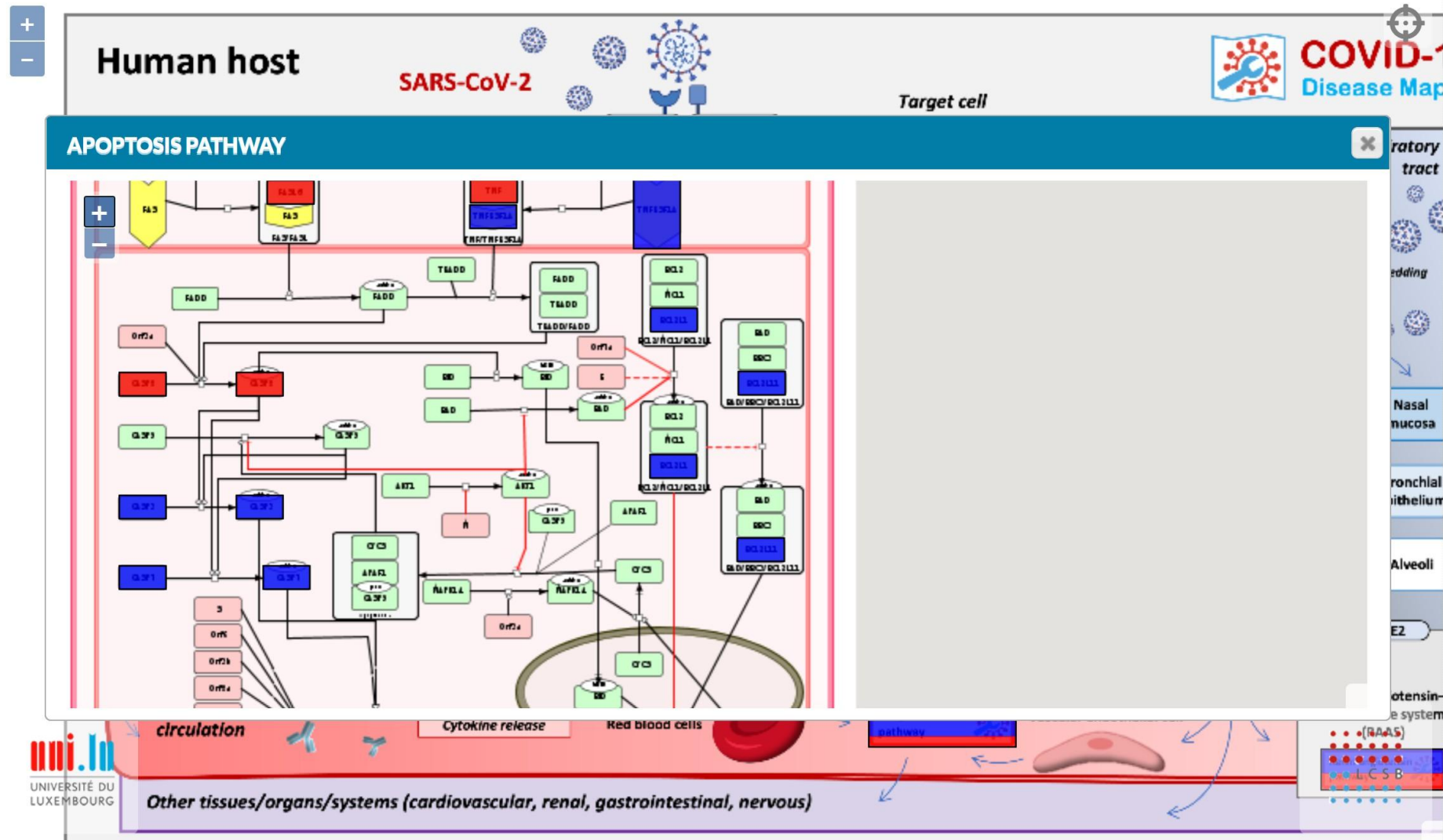
Many different modelling approaches – one common language?



- A format to encode mathematical models that is used in systems biology.
- Initially focused on non-spatial, reaction-based biochemical models.
- Packages covering different modelling approaches (qual).
- Supported by software libraries in different programming languages .
- Can be imported or exported by a variety of modelling and simulation tools.
- Does not store experimental data, or simulation descriptions.
- Based on xml format, intended to be machine and not human-readable.



Name	View
Apoptosis pathway 	
Coagulation pathway 	
E protein interactions 	
Electron Transport Chain disruption 	
Endoplasmatic Reticulum stress 	
HMOX1 pathway 	
Interferon 1 pathway 	



The COVID-19 Disease Map

<https://covid19map.elixir-luxembourg.org/minerva/>

The content

	Source		
	Individual diagrams	Reactome	WikiPathways
Diagram contents	21 diagrams 1334 interactions 4272 molecular entities 397 publications	2 diagrams 101 interactions 489 molecular entities 227 publications	19 diagrams 401 interactions 738 molecular entities 61 publications
Access	Gitlab git-r3lab.uni.lu/covid/models	SARS-CoV-1 and SARS-CoV-2 infections collection reactome.org/PathwayBrowser/#/R-HSA-9679506	COVID pathways collection covid.wikipathways.org
Exploration	The MINERVA Platform [21] covid19map.elixir-luxembourg.org Guide: covid.pages.uni.lu/minerva-guide	Native web interface Guide: covid.pages.uni.lu/reactome-guide	Native web interface Guide: covid.pages.uni.lu/wikipathways-guide
Biocuration guidelines	Community ⁴	Community ⁵ Platform-specific ⁵	Community ⁶ Platform-specific ⁶
Diagram Editors	CellDesigner ⁷ , Newt ⁸ SBGN-ED [22], yEd+ySBGN ⁹	Reactome pathway editor ⁵	PathVisio [23]
Formats	CellDesigner SBML [24] SBGNML [25,26]	Internal, SBML and SBGNML compliant	GPML [23]

Article | 19 October 2021 | [OPEN ACCESS](#)

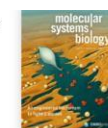
[TRANSPARENT PROCESS](#)

COVID19 Disease Map, a computational knowledge repository of virus–host interaction mechanisms

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Author Information

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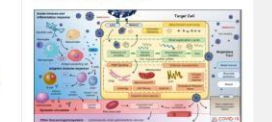


The main groups of the C19DMap Community are biocurators, domain experts, and analysts and modelers, communicating to refine, interpret, and apply C19DMap diagrams. These diagrams are created and maintained by...

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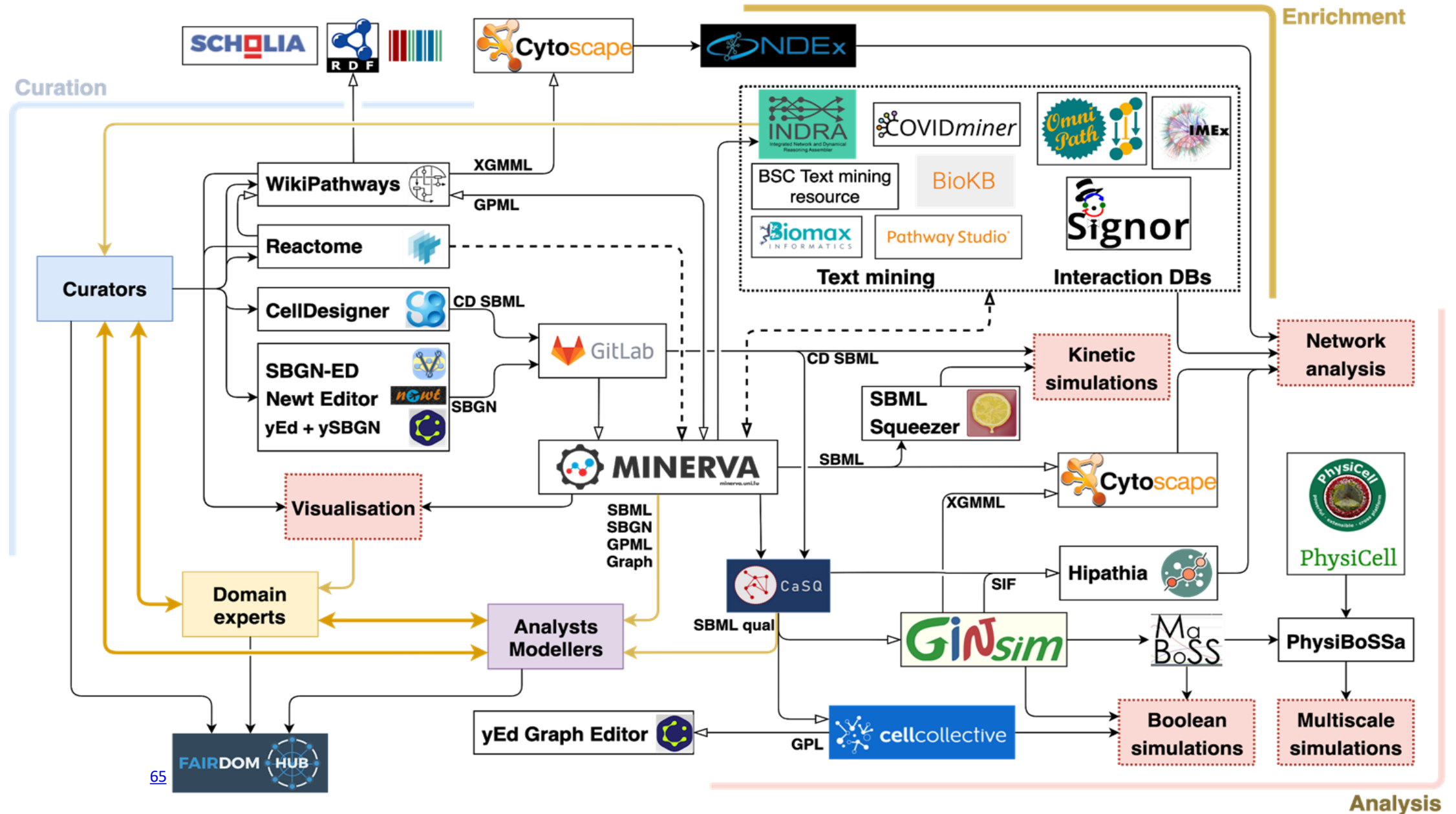
Figure 2.

The structure and content of the COVID-19 Disease Map



An overview of the areas of focus of the C19DMap biocuration. Go to

The COVID19 Disease Map ecosystem



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ORIGINAL RESEARCH article
Front. Immunol., 13 February 2024
Sec. Systems Immunology
Volume 14 - 2023 | <https://doi.org/10.3389/fimmu.2023.1282869>

Drug-target identification in COVID-19 disease mechanisms using computational systems biology approaches

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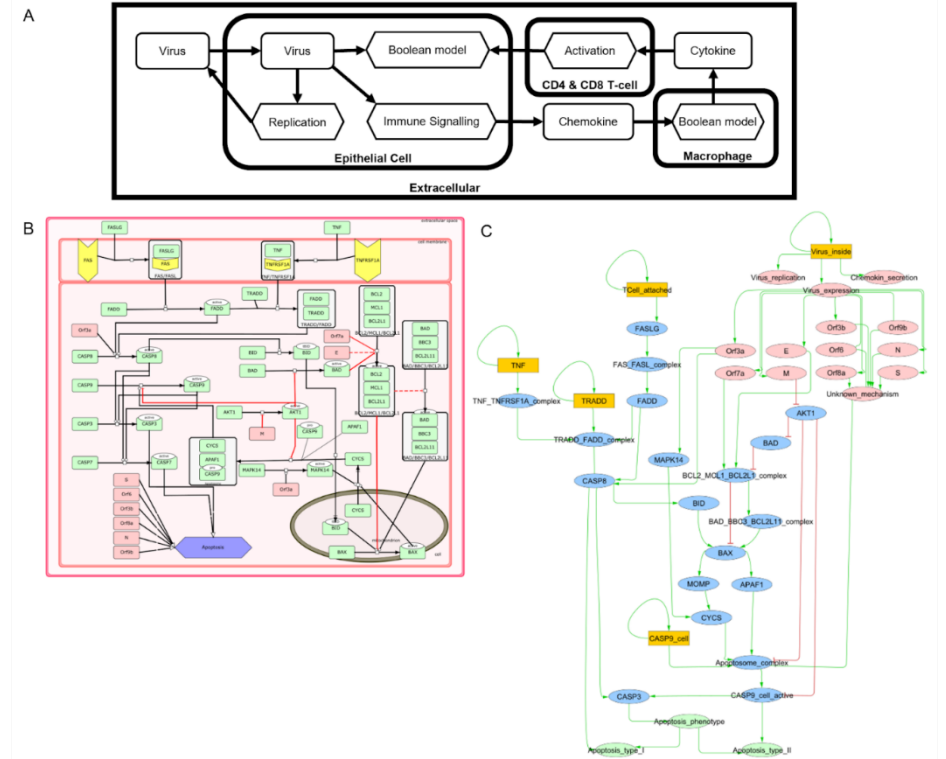
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Data availability statement
Ethics statement
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Funding
Acknowledgments



An example of simulations

Simulation of wild type and mutants using PhysiBoSS. Our framework can simulate wild-type epithelial cells state (A)

and wild-type immune cells recruitment (B)

and study the effect of knockouts such as FADD's in the epithelial cells' apoptosis (C)

or such as p38's in the immune cells recruitment (D).

